

Unequal Mobilization: Military Loss, Federal War Investment, and the Geography of Postwar Development

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Abstract

World War II removed men from particular American counties and placed industrial capital in others. This paper asks whether the two coincided. Assembling a 3,073-county panel from the Civilian Production Administration’s contract listing, the War Production Board’s facilities report, 8.3 million Army enlistment records, and the Haines census series, I find that the wartime state made three distinct allocations — procurement contracts, publicly financed industrial plant, and military installations — whose pairwise overlap runs from 0.22 to 0.59, against 0.87 for deaths and population on the same scale. None tracked sacrifice. Per resident on the same counties, contract dollars correlate with the 1940 manufacturing share at 0.69, their strongest tie to anything prewar, with the urban share at 0.65, and with fatal burden at -0.03 ; plant and installations behave the same way, at $+0.01$ and -0.05 . The allocation function loaded heavily on prewar industrial capacity and negligibly on human loss, and at this sample size all three exclude any association above 0.09 in either direction. In matched comparisons the investment that arrived left durable and divergent structures: through 1970 a war plant raised county manufacturing employment by 39 log points, an industrial workforce 47 percent larger, faster than it raised employment as a whole, while a military installation raised total employment by more — 41 log points against 22 — and did not measurably raise manufacturing employment. The installation also

*University of Chicago. Email: Joeblum@uchicago.edu. Replication code and the assembled county panel are available from the author. The contract microdata are the Brunet–Koustas digitization of the Civilian Production Administration listing, distributed by the authors (Brunet et al., 2025c), and the plant microdata are Garin and Rothbaum’s digitization of the 1945 War Production Board facilities report; casualty and enlistment records are ICPSR 38927; county outcomes are ICPSR 2896. The county assignment, harmonization and all estimates are my own, as are all errors.

lowered measured homeownership by seven and a half points, part of which is the tenure composition a standing garrison imposes. With respect to sacrifice the war state was not hostile but indifferent.

JEL classification: N42, N92, H56, R11, J15.

1 Introduction

World War II is usually narrated at the scale of the nation. More than sixteen million Americans served in the armed forces; factories were converted into an “arsenal of democracy”; households paid new taxes, rationed consumption, purchased war bonds, and entered war industries. At that scale, mobilization appears as a unified national experience. Yet the war was administered through places. Men left particular counties. Death notices returned to particular counties. Contracts, shipyards, aircraft factories, ordnance works, machine tools, and publicly financed plants were concentrated in particular counties. The wartime state therefore produced two enormous local shocks at once: it removed people from communities through military service and death while injecting demand and productive capital into selected industrial spaces.

This paper asks whether those two geographies coincided. The primary question is empirical:

How geographically aligned were military sacrifice and federal economic investment during World War II, and did the divergence between casualty burden and wartime spending predict which American communities emerged from the war with durable economic advantages?

A second question follows from the first and cannot be answered by the same methods:

How does that uneven geography complicate wartime political representations of sacrifice as nationally shared?

The two are not independent. The rhetorical claim of shared sacrifice was a claim *about* distribution, made while the distribution was being decided. Measuring the geography therefore does not merely supply background for the rhetoric; it establishes what the rhetoric was competing against. The empirical sections answer the first question; Section 6 answers the second. The question is deliberately different from asking whether World War II “helped” the American economy. It asks instead whether the human costs and productive assets of mobilization were distributed across the same places.

The distinction matters because the two literatures that make this question possible have developed largely in parallel. Economic historians show that wartime spending altered local development, but not uniformly. Fishback and Cullen find substantial population effects with comparatively limited per-capita gains from aggregate wartime spending (Fishback and Cullen, 2013). Jaworski identifies wartime mobilization as a force in the industrialization of the South (Jaworski, 2017). Garin and Rothbaum show that large publicly financed plants

generated strikingly persistent local effects: by 1970, treatment counties had substantially more manufacturing employment, larger populations, and higher family incomes than comparable places (Garin and Rothbaum, 2025). Recent work also emphasizes that the composition of mobilization spending matters for long-run growth (Jaworski and Yang, 2025; Brunet, 2026).

A second literature studies mobilization as a labor and demographic shock. Acemoglu, Autor, and Lyle exploit cross-state variation in male military mobilization to study female labor supply and wages (Acemoglu et al., 2004). Brodeur and Kattan show that county-level wartime casualties reduced the size of the baby boom and were associated with higher female employment and household income in the 1950s (Brodeur and Kattan, 2022). Ferrara demonstrates that casualties among semiskilled White soldiers contributed materially to Black occupational upgrading at midcentury (Ferrara, 2022). These findings establish that wartime death was not merely a demographic subtraction; it changed local labor-market structure and the distribution of opportunity.

A third literature, largely political-science, asks the distributional question directly. Kriner and Shen (2010) document a widening “casualty gap” across American communities from Korea through Iraq, and argue that the concentration of military death in poorer places has measurable consequences for political participation and for public tolerance of war. Their object is the social gradient of sacrifice within the casualty distribution. The object here is different and complementary: not who bore the loss, but whether the state’s simultaneous placement of productive capital fell in the same places. A casualty gap and an investment gap are separate facts about the same war, and this paper measures the correspondence between them.

The contribution of this paper is to place these shocks in one empirical design. The central object is not a “return on sacrifice” ratio, which would be statistically unstable and morally incoherent, but the spatial relationship between two national distributions. I construct county-level measures of fatal military exposure and wartime industrial investment, estimate their geographic overlap, and model their separate and interactive association with postwar development. The empirical strategy distinguishes temporary contracts from durable facilities, uses internally consistent Army/Army Air Forces casualty and enlistment universes for causal work, and treats broader all-service casualty maps as descriptive because the underlying historical lists have non-identical definitions.

The paper also contributes to the history of wartime political language. Roosevelt’s 1942 fireside chat described taxes, wage restraint, rationing, bond purchases, factory work, and military service within an “economy and equality of sacrifice” (Roosevelt, 1942). The Office of War Information elaborated the same moral grammar visually, teaching civilians to imagine consumption restraint, war work, and bond buying as extensions of battlefield service

(Winkler, 1978; Sparrow, 2011). Quantitative disaggregation therefore tests not whether the rhetoric was sincere, but what it compressed. A language of nationally shared sacrifice may coexist with a highly unequal geography of both loss and productive inheritance.

This paper’s findings can be stated in advance. First, the wartime state made not one allocation but three, and they do not coincide: procurement contracts, publicly financed industrial plant, and military installations occupy geographies whose pairwise overlap runs from 0.22 to 0.59, and of 2,973 counties only two received both a large public plant and a top-decile installation, against 2.2 expected under independence. Second, none of the three tracked sacrifice, and the cleanest way to see it is to put the allocations next to the characteristics the procurement agencies were selecting on. Per resident, contract dollars correlate with the 1940 manufacturing share at +0.693, the urban share at +0.648 and log population at +0.666; with fatal burden they correlate at -0.026 . Plant and installations load on the same prewar characteristics at +0.23 to +0.51 and on burden at +0.013 and -0.054 . These near-zeros are precisely estimated — the confidence intervals exclude any association above 0.09 in absolute value, and the one that does exclude zero, -0.054 for installations, is negative — and they turn negative when deaths are conditioned on enlistment. Coverage tells the same story: 98.2 percent of counties recorded a death while 48.5 percent received any major contract. Third, the investment that did arrive left durable and divergent structures. In matched comparisons, a large new publicly financed plant raised county population, manufacturing employment and family income through 1970, and raised manufacturing employment faster than employment as a whole. A military installation raised population and total employment by more, left manufacturing employment where it found it, and lowered homeownership by seven and a half points.

The paper proceeds as follows. Section 2 frames mobilization as a spatial allocation problem. Section 3 describes the sources and the construction of the county panel. Section 4 sets out the alignment statistics and the estimating equations. Section 5 reports the results, including two matched designs and a systematic demonstration that no construction of the natural instrument satisfies its exclusion restriction. Section 6 reads the measured geography against the wartime rhetoric of shared sacrifice. Section 7 concludes with an explicit statement of what the evidence does and does not support.

Three claims organize the paper. First, “war mobilization” is not one local treatment. Fatal exposure, procurement demand, and durable public capital are analytically distinct shocks. Second, the correct quantity of interest is correspondence rather than compensation: the spatial overlap between the distribution of human loss and the distribution of productive investment. Third, the political language of shared sacrifice is historically revealing precisely because it translated heterogeneous local experiences into a common national idiom.

2 War Mobilization as a Spatial Allocation Problem

The scale of American mobilization makes its local heterogeneity easy to obscure. Official Department of Veterans Affairs statistics report 291,557 battle deaths and 113,842 other deaths in service during World War II (Department of Veterans Affairs, 2025). The newly digitized ICPSR archive contains 300,131 Army/Army Air Forces fallen, 65,507 Navy/Marine Corps/Coast Guard casualty-list observations, and 8,293,187 Army/AAF enlistment records (Ferrara, 2024). These records make it possible to reconstruct military exposure at county scale, but they also reveal an important measurement constraint: the two casualty lists do not share a common universe, and each source documents its own. DS1 is the War Department’s *Honor List of Dead and Missing* (War Department, 1946), and its status field sorts 299,947 of the 300,131 records into six categories: killed in action (171,530 records, 57.2 percent), *died non-battle* (82,479, 27.5 percent), died of wounds (24,786, 8.3 percent), finding of death (18,888, 6.3 percent), died of fatal battle injury (912, 0.3 percent), and missing (1,352, 0.5 percent); the remaining 184 records carry no status code (Ferrara, 2024). More than a quarter of the Army list is therefore death outside combat — sickness, accident, homicide or suicide in a line-of-duty status (Congressional Research Service, n.d.). DS3 comes from a different document, the *State Summary of War Casualties* compiled service by service and state by state (Department of the Navy, 1946), which is named in the DS3 entries themselves.¹ Its published tables report the dead in two categories only, combat and prison camp, with no non-battle column at all (Congressional Research Service, n.d.). Adding a list in which 27.5 percent of entries are non-battle deaths to one that does not record them would manufacture comparability the sources do not possess, and would do so unevenly across states.

The geography of wartime industrialization was no less uneven. Two wartime agencies recorded it at the firm and plant level: the Civilian Production Administration listed major supply contracts (Civilian Production Administration, 1946a), and the War Production Board listed authorized manufacturing facilities (War Production Board, 1945). In the legacy county series used by Fishback and Cullen, roughly two-fifths of major wartime spending accrued to the top twenty counties (Fishback and Cullen, 2013). Working from the record level rather than from any published county aggregate — the contract file is the Brunet–Koustas digitization of the four CPA volumes (Brunet, 2017), to which I add the county assignment — yields 190,693 contract records. Aggregated to counties it recovers \$178.2 billion, of which \$177.3 billion falls in the 3,070 counties the County Data Book also covers, against that series’ \$180.4 billion; it correlates with that series at 0.992 in levels, and places 44.0 percent

¹Several DS3 records carry the reference verbatim in their transcribed text — “Published Missing on Maine State Summary of War Casualties,” “Carried Missing on Colorado State Summary” — which is how the source document can be identified from the data alone.

of dollars in the top twenty counties against the older series’ 42.7 percent. It also records 1,583 counties with no major contract against the County Data Book’s 1,506, implying that the legacy county measure contains nontrivial noise among places with little or no major federal production spending.

This concentration has a theoretical implication. If productive wartime investment created persistent agglomeration, industrial learning, housing, infrastructure, and labor-market depth, then the postwar economic legacy of mobilization depended not only on how much the federal government spent nationally but on where durable capital remained after military demand collapsed. Garin and Rothbaum’s estimates strongly support this distinction: large public wartime plants generated long-run local increases in manufacturing employment, population, family income, and incumbent earnings (Garin and Rothbaum, 2025). Aggregate procurement flows, by contrast, produced more limited per-capita persistence in the Fishback–Cullen results (Fishback and Cullen, 2013). Accordingly, this paper treats wartime investment as a vector rather than a scalar.

The same logic applies to military loss. A county’s fatality rate can alter family formation, labor supply, occupational substitution, and the local distribution of opportunity even if it does not change aggregate county income in one direction. Brodeur and Kattan’s fertility and female-employment results and Ferrara’s occupational-upgrading results make a single signed “casualty effect” theoretically implausible (Brodeur and Kattan, 2022; Ferrara, 2022). Military loss is thus neither simply a demographic denominator nor an ethical counterweight to spending. It is a local shock with its own mechanisms.

3 Data

The analysis assembles four archival and administrative sources into a single county panel. Table 1 states the scale of each; the combined file rests on roughly

8.6 million individual military records — the 300,131-name Army honor list and the 8,293,187-record enlistment file — and 205,000 individual wartime contract and plant records, aggregated to 3,073 counties.

Military exposure. The ICPSR World War II archive digitizes the War Department honor list and the NARA enlistment file (Ferrara, 2024; War Department, 1946). DS1 supplies 300,131 Army and Army Air Forces fatalities with state and county of residence; the county identifier is the county FIPS code, which I verify against the 1940 census universe. Reaching the panel from the honor list costs 0.8 percent of it, and the whole of that loss is accounted for here. Of the 300,131 records, 299,416 (99.76 percent) carry a resolvable county; 715 do not. Among those that do, 297,887 hold one of the five statuses that denote a death and

Table 1: Primary sources and their scale

Source	Unit of record	Records	Supplies
ICPSR 38927 DS1	individual soldier	300,131	Army/AAF fatalities, D_i^A
ICPSR 38927 DS2	individual enlistee	8,293,187	soldier composition
Ferrara county tabulation	county	3,073	enlistment denominator, E_i^A
ICPSR 38927 DS3	individual casualty	65,507	Navy/USMC/USCG (descriptive only)
CPA contracts	individual contract	190,693	procurement, W_i^C
WPB facilities	individual plant	14,681	war plant, W_i^F, G_i
Haines ICPSR 2896	county $\times$ year	8 parts	outcomes and controls
Fishback–Horrace–Kantor	county	3,067	prewar controls, coordinates

Notes: DS3 is reported for completeness and is used nowhere in estimation: the Navy, Marine Corps, and Coast Guard summaries exclude deaths inside the United States and deaths from disease, homicide, or suicide, so they are not additive with the Army honor list. All estimation is conducted inside the internally consistent Army/AAF universe.

are counted in D_i^A , while 1,348 are carried as missing and 181 have no status. A further 260 fall in county codes with no counterpart on 1940 boundaries. What enters the panel is therefore 297,627 Army/AAF deaths, of which 195,677 are battle deaths. DS2 supplies 8,293,187 individual enlistment records. Its geography is coded in a NARA administrative scheme rather than FIPS, and I decode it from the NARA county code list: 7,280,828 records (87.8 percent) resolve to a 1940 county, and 99.8 percent of the derived county codes fall inside the census universe. The residual is dominated by records coded “United States at large,” Alaska, Hawaii, the Panama Canal Zone, and foreign residence.

The enlistment denominator itself is not built from DS2. E_i^A is Ferrara’s county tabulation of draftees and volunteers in the same NARA universe, obtained through the Garin and Rothbaum (2025) replication package (Ferrara, 2022); it totals 6.94 million men and is what the per-enlistee rates in this paper divide by. County totals built from my own DS2 decode reach 7.19 million and correlate with that tabulation at 0.990, which is reported as a check on the decode rather than as the denominator. What DS2 supplies that nothing else can is the individual record: race, education, birth year and enlistment source for all 8.29 million, and the Army General Classification Test score for the 447,464 records — 5.4 percent — that carry one. Those fields are what make the composition adjustment of Section 5 possible. DS2 also carries a killed-in-service flag, set on 234,386 of its records. That flag is not used as a death measure anywhere in this paper and should not be read as a competing count: it marks enlistment records that were later annotated, so its universe is whoever appears in the enlistment file, while the honor list is a record of the dead compiled to be complete. The two are not alternative measurements of the same quantity, and only the honor list is treated as one.

Wartime investment. Two wartime records supply it, and they are different documents produced by different agencies. Procurement comes from the Civilian Production Administration’s cumulative alphabetical listing of major war supply contracts, June 1940 through September 1945 (Civilian Production Administration, 1946a); durable capital comes from the War Production Board’s report on manufacturing facilities authorized through October 1944 (War Production Board, 1945). I work from the record level of each rather than from any published county aggregate. The contract file yields 190,693 obligations worth \$181.0 billion; the facilities file yields 14,681 plant records worth \$20.8 billion, of which 76.6 percent was publicly financed.² Neither record gives a county, only a place, and Section 5 reports how far the place-to-county assignment succeeds and how it validates against the contemporaneous official series.

Neither file was assembled by me. The contract volumes were transcribed to machine-readable form by E-Records USA and cleaned by Gillian Brunet and Dmitri Koustas (Brunet, 2017); the copy used here is the record-level file the authors distribute from their research page (Brunet et al., 2025c). It holds 190,693 records against the 190,599 of the dissertation’s own analysis sample, so it is a later vintage of that pipeline rather than the identical extract.³ The facilities report was photographed at the Harvard University Library and digitized by Garin and Rothbaum (Garin and Rothbaum, 2025; War Production Board, 1945). What is constructed here is the place-to-county assignment, the independent-city merge, the county aggregation, and everything downstream of them.

County outcomes and controls. Decennial census counts, the County Data Books, and the prewar control vector come from the Haines historical county database and the Fishback–Horrace–Kantor county file (Haines and ICPSR, 2010; Fishback et al., 2005). The panel is built on 1940 boundaries, with population-weighted crosswalks available for every census year (Ferrara et al., 2024).

Sample. The baseline sample is the 3,073 continental counties observed on 1940 boundaries, holding 131.7 million residents. Alaska, Hawaii, and the territories are excluded: statehood

²The facilities source reports government-operated plants in a hierarchy — a national total, then eight department subtotals, then the individual arsenals and yards — which a flat read of the sheet counts three times. The eight department figures sum to exactly the national total above them, which is how the structure was identified. Removing the aggregate rows, twelve in all once the three departments the parser emitted twice are counted, takes the volume from \$24.0 billion to \$20.8 billion. Six of them had acquired the location of the facility printed beneath them and so reached the county panel as phantom plants, \$307 million of it in Colbert County, Alabama. Correcting this changes three counties and no qualitative result; the affected figures throughout are the corrected ones.

³The differences are confined to record selection and date imputation — 182 contracts whose recorded completion precedes their award survive here, worth 0.09 percent of value, and 15,382 carry an imputed completion year against the roughly one hundred the dissertation describes. This paper uses no contract timing: W_i^C is a cross-section of total obligated dollars, so neither difference touches any estimate.

timing and administrative comparability make them non-analogous.⁴ Independent cities are merged into their surrounding county. The merge is not a cosmetic adjustment. The wartime sources split them — the Civilian Production Administration recorded contracts at the city while the Army recorded residence at the county — leaving fourteen units that between them held \$6.82 billion in contracts and 2.26 million residents but only 251 recorded deaths, a rate of 0.11 per thousand against a national 2.26. Left unmerged these manufacture high-investment, zero-sacrifice counties and deflate every alignment statistic in the paper. Parent counties are identified from the 1940 boundary file by longest shared border.

Provenance. Three of the sources above are prior authors’ replication archives, and it is worth stating exactly what each supplies so that the boundary between constructed and inherited quantities is not left to inference. From Garin and Rothbaum (2025) this paper takes their digitization of the War Production Board facilities report (War Production Board, 1945), the 1910 state ICPSR–FIPS crosswalk, a city-to-county crosswalk used as one input to the geocoder, the NBER county adjacency file used for the neighbor comparison, the Fishback–Horrace–Kantor file of county geographic and infrastructure characteristics (Fishback et al., 2005), Ferrara’s county tabulation of draftees and volunteers, which supplies the enlistment denominator E_i^A (Ferrara, 2022), and a metropolitan-status file that proved unusable on this panel — seven files in all. From Jaworski (2017) it takes the county counts of establishments allocated under the 1938 Industrial Mobilization Plan, the sole basis for the instruments in Section 5.9. From the Brunet–Koustas project (Brunet, 2017) it takes the record-level contract file. What is constructed here is the place-to-county assignment for both wartime records, the independent-city merge, the wartime exposure measures, the enlistment-composition variables, the alignment statistics, and both matched designs. No estimate in this paper reuses a treatment effect, standard error, or matched sample from a prior archive.

The paper uses two measures of military burden because they answer different questions. Community loss is defined as

$$B_i^{\text{civic}} = 1000 \frac{D_i}{P_{i,1940}}, \quad (1)$$

where D_i is attributable wartime military death and $P_{i,1940}$ is county population. This is the appropriate measure for descriptive maps because it captures the loss borne by the community

⁴Alaska and Hawaii were territories throughout the war and did not enter the County Data Book series on the same basis as the states. Hawaii in particular would be a severe outlier on every investment measure, and its wartime administration under martial law from December 1941 has no continental analogue.

as a whole. For inference, the preferred measure remains within the internally consistent Army/AAF universe:

$$B_i^A = 1000 \frac{D_i^A}{E_i^A}, \quad (2)$$

where D_i^A denotes Army/AAF fallen and E_i^A Army/AAF enlistment records. A secondary specification uses military-age men in the 1940 census as the denominator. The distinction prevents differences in county age–sex composition from being misread as differences in fatality intensity among those actually exposed to service.

E_i^A is not an exogenous denominator, and the paper does not treat it as one. Wartime manpower was administered through some 6,400 local Selective Service boards, each staffed by unpaid civilian volunteers with substantial discretion over classification, deferment, and the order of induction (Selective Service System, 1947; Flynn, 1993). Boards deferred agricultural and war-production labor at rates that varied with the local economy, so a county’s enlistment count reflects its industrial structure as well as its population.⁵ Angrist and Krueger (1994) show that WWII veteran status is selected on characteristics that independently predict earnings, and that comparisons of veterans to nonveterans are biased by that selection; Bound and Turner (2002) exploit the changing manpower requirements of the period for the same reason. Both results bear on E_i^A directly: it is the output of a screening process correlated with prewar local characteristics, which is why the paper conditions on the full prewar control vector wherever B_i^A appears and reports B_i^{civic} alongside it throughout.

B_i^A is defined only where the denominator is credible. A county fails the screen if its enlistment count exceeds its 1940 male population aged 14 to 45, if the implied fatality rate exceeds 300 deaths per 1,000 enlistees, or if either quantity is missing — all three being signatures of an enlistment record whose county of residence is unreliable rather than of a county that suffered extraordinarily. Ninety-seven of the 3,073 counties fail it, holding 2.55 percent of the 1940 population. Every quantity below that is expressed per enlistee is computed on the surviving 2,976; every quantity expressed per resident uses all 3,073.

Wartime investment is decomposed into procurement and durable capital. Let

$$W_i^C = \text{arsinh} \left(\frac{\text{major contract dollars}_i}{P_{i,1940}} \right), \quad (3)$$

$$W_i^F = \text{arsinh} \left(\frac{\text{facility dollars}_i}{P_{i,1940}} \right), \quad (4)$$

⁵The direction of the resulting bias works against the paper’s finding rather than for it. If war-production counties deferred more men, their enlistment denominators are smaller and their measured per-enlistee fatality rates correspondingly higher, which would push the correlation between burden and investment upward. The estimated correlation is nonetheless approximately zero, and turns negative under composition adjustment.

with an additional treatment indicator G_i for large publicly financed wartime plants in specifications designed to parallel Garin and Rothbaum (2025). The inverse-hyperbolic-sine transformation preserves true zeros while limiting the leverage of extreme spending values.⁶ The reconstructed contract series is preferred; the County Data Book measure enters as a robustness check rather than as unquestioned ground truth.

The primary outcome vector is deliberately multidimensional:

$$\mathbf{Y}_{it} = \left(\log P_{it}, M_{it}, \log E_{it}^M, H_{it}, \log y_{it} \right), \quad (5)$$

where P is population, M the manufacturing employment share, E^M manufacturing employment in levels, H homeownership, and y household or family income. This prevents “postwar advantage” from collapsing migration, industrialization, wealth accumulation, and distributional change into one index. The share and the level are both carried because they can move in opposite directions: a treatment that raises every sector raises the level of manufacturing employment while lowering its share, and Section 5.8 shows that a military installation does exactly that.

⁶The choice matters here more than it usually does: 1,583 counties received no major contract and 2,412 received no publicly financed plant, so a log transformation would discard more than half the sample on one measure and more than three-quarters on the other. Coefficients on an arsinh-transformed regressor approximate elasticities away from zero but not near it, and are read as semi-elasticities throughout.

4 Empirical Strategy

The first estimand is descriptive but conceptually central. Let each county's share of national military deaths and national wartime investment be

$$p_i^D = \frac{D_i}{\sum_j D_j}, \quad p_i^W = \frac{W_i}{\sum_j W_j}. \quad (6)$$

I measure geographic alignment with the overlap coefficient

$$\mathcal{A} = \sum_i \min(p_i^D, p_i^W) = 1 - \frac{1}{2} \sum_i |p_i^D - p_i^W|, \quad 0 \leq \mathcal{A} \leq 1. \quad (7)$$

An overlap of one implies identical county shares in the two national distributions; zero implies disjoint geographic mass. Unlike a ratio of dollars to deaths, $\mathcal{A}$ measures correspondence without monetizing human loss.⁷ Spearman rank correlation and Jensen–Shannon divergence provide complementary measures of ordinal association and distributional distance:

$$\text{JS}(p^D, p^W) = \frac{1}{2} \text{KL}(p^D \| m) + \frac{1}{2} \text{KL}(p^W \| m), \quad m = \frac{1}{2} (p^D + p^W). \quad (8)$$

For visualization, both wartime exposures are residualized on a common set of prewar characteristics X_i : 1940 log population, the urban share, the Black population share, the manufacturing share, and 1930–40 population growth. Writing $\tilde{B}_i$ and $\tilde{W}_i$ for those residuals, the conditional imbalance measure is

$$I_i = z(\tilde{B}_i) - z(\tilde{W}_i). \quad (9)$$

Positive values identify casualty-heavy counties conditional on observables; negative values identify investment-heavy counties. This scalar is used for maps and descriptive typologies only. The causal analysis preserves the two treatment dimensions separately.

For $t \in \{1950, 1960, 1970\}$, the baseline long-difference specification is

$$\begin{aligned} \Delta Y_{i,1940 \rightarrow t} = & \beta_C W_i^C + \beta_F W_i^F + \beta_B B_i^A + \theta_C (W_i^C \times B_i^A) \\ & + \theta_F (W_i^F \times B_i^A) + X_i' \gamma + \delta_s + \varepsilon_i. \end{aligned} \quad (10)$$

⁷ $\mathcal{A}$ is bounded in $[0, 1]$, is invariant to the national totals of D and W , and equals one only when the two county share vectors are identical. It is the complement of half the total variation distance, which is why the second equality in its definition above holds. Its weakness is insensitivity to the ordering of counties, which is why the Spearman correlation is reported alongside it.

The interaction coefficients ask whether the postwar association of investment differs with local fatality exposure: θ_C for temporary procurement, θ_F for durable facility investment. They are reported in Table A5. A dynamic extension carries the pre-trend test back to 1920, before the Depression as well as before the war (Table A4). Standard errors are clustered by state and checked against spatial-HAC inference.

Equation (10) is not, by itself, a causal design. Contract placement reflected prewar industrial capacity, transportation infrastructure, military procurement priorities, and political economy. The 1938 Industrial Mobilization Plan may therefore be used as a predetermined predictor or instrument only where its exclusion restriction can be defended explicitly (Fishback and Cullen, 2013; Jaworski, 2017). A stronger quasi-experimental exercise isolates large publicly financed plants whose placement outside major industrial centers was shaped partly by wartime security considerations, following Garin and Rothbaum (2025). The casualty dimension is likewise subjected to composition adjustment: individual enlistment characteristics are used to construct predicted fatality risk, allowing observed county deaths to be decomposed into expected and excess components.

The inferential hierarchy is therefore deliberate. National overlap and residualized imbalance are descriptive. Equation (10) estimates conditional associations. Causal interpretation is reserved for narrower specifications in which assignment to wartime capital or fatality risk is plausibly quasi-exogenous.

5 Results

The county merge described in Section 3 is executed on 3,073 continental counties observed on 1940 boundaries. Table 2 reports the estimation sample.

Before turning to alignment, two construction results establish that the wartime measures are what they claim to be.

5.1 Validating the wartime measures

The contract series is reconstructed from the Civilian Production Administration volumes and assigned to counties through a historical place crosswalk; 98.5 percent of contract dollars resolve to a county; Table A1 decomposes that figure by match stage, so every assignment is attributable to a named rule. Table 3 compares the result against the 1947 County Data Book (Bureau of the Census, 1947), the contemporaneous official series used by Fishback and Cullen (2013), as distributed in Part 70 of the Haines archive (Haines and ICPSR, 2010). On the 3,070 counties both series cover, national totals differ by 1.7 percent, the levels correlation is

Table 2: Estimation sample: 3,073 continental U.S. counties on 1940 boundaries

Variable	Mean	SD	p10	Median	p90	<i>N</i>
<i>Wartime exposure</i>						
Army/AAF deaths	97	315	11	43	157	3,073
Fatal burden B_i (per 1,000 residents)	2.74	9.26	1.35	2.30	3.63	3,073
War supply contracts (\$000s)	57,980	406,004	0	0	43,911	3,073
Publicly financed war plant (\$000s)	4,882	26,995	0	0	3,891	3,073
Military installations (\$000s)	3,197	12,214	0	0	7,903	3,073
<i>Prewar characteristics, 1940</i>						
Population	42,847	145,241	5,822	18,698	69,078	3,073
Urban share	0.229	0.247	0.000	0.181	0.597	3,073
Black population share	0.106	0.178	0.000	0.012	0.394	3,073
Manufacturing employment share	0.028	0.038	0.000	0.012	0.080	3,073
Log population growth 1930–40	0.051	0.148	-0.106	0.045	0.199	3,073
<i>Postwar outcomes</i>						
Population 1950	49,043	171,132	5,556	18,734	80,602	3,072
Population 1970	65,548	232,065	5,014	18,780	111,825	3,065
Manufacturing share 1960	18.85	12.90	3.20	17.10	37.80	3,071
Manufacturing share 1970	21.85	13.57	4.10	21.00	41.00	3,066
Median family income 1950 (\$)	2,240	865	1,108	2,311	3,319	3,071
Median family income 1970 (\$)	7,460	1,853	5,158	7,398	9,816	3,066
Homeownership rate 1970 (%)	70.9	8.1	60.6	72.0	79.9	3,066

Notes: Wartime exposure and prewar characteristics as constructed in Section 3. Contract and facility dollars are in thousands of current dollars. Fatal burden is Army/AAF deaths per 1,000 1940 residents.

0.992, and the top-twenty concentration is close to the older measure’s own (44.0 against 42.7 percent). The two series were built from different documents through different geographies, so the agreement is informative rather than mechanical.

Table 3: Reconstructed wartime series against the 1947 County Data Book

Series	Reconstructed (\$bn)	County Data Book (\$bn)	Pearson (levels)	Pearson (log1+x)	Spearman (rank)
Major war supply contracts	177.3	180.4	0.992	0.959	0.957
War plant, publicly financed	15.0	16.7	0.969	0.899	0.862

Notes: The County Data Book series is the contemporaneous official measure used by Fishback and Cullen (2013). The facilities comparison uses the County Data Book’s *industrial* series only; its military series is treated as a separate allocation throughout. Correlations across the 3,070 counties carried by both series.

The facilities report yields 14,681 plant records totalling \$20.8 billion, of which 76.6 percent was publicly financed. Two limitations must be stated plainly. The first is a coverage cutoff: the War Production Board compiled the volume in March 1945 and it records facilities authorized *through October 1944*, so whatever was authorized in the last ten months of the war is absent. Nothing in the source permits a size for that omission, and I do not offer one;

what can be said is that new authorization was already far past its peak, with the availability dates in the volume falling from \$7.0 billion of public plant in 1943 to \$4.6 billion in 1944. Plants missing on this account enter the panel as untreated, which attenuates the estimates of Section 5.7 toward zero rather than inflating them. The second limitation is unassignability: 180 facilities holding \$0.59 billion—3.7 percent of all publicly financed plant dollars—are recorded by the source with no place, credited either to “various” locations or to a company whose listing continues from the previous page. They are unassignable in principle. What this paper calls W_i^F is therefore *locatable* publicly financed plant investment, and 97.8 percent of the assignable public dollars resolve to a county, landing in 661 of them. Against the County Data Book’s separate industrial-facilities series the correlation is 0.969.

5.2 Three allocations, not two

The paper’s framework anticipated two forms of wartime investment. The 1947 County Data Book distinguishes a third. Its facilities series separates *industrial* projects from *military* ones—camps, airfields, depots, proving grounds—and the two select largely different counties (Spearman $\rho = 0.31$). Of the 1,010 counties receiving any war facility, 386 received publicly financed plant only (W^F), 349 a military installation only (W^M), and 275 both. (The County Data Book’s own two columns partition a slightly wider set, 1,059 counties, at 435/334/290; the paper uses W^F throughout so that this split and the regressions describe the same measure.) Table A2 reports how those three groups differ on prewar and postwar characteristics. Table 4 reports the pairwise geography of all four wartime allocations.

Table 4: Pairwise geography of the four wartime allocations

Pair	Overlap $\mathcal{A}$	JS (bits)	Spearman ρ
$W^C \times W^F$	0.591	0.215	0.616
$W^C \times W^M$	0.216	0.647	0.343
$W^C \times D$	0.466	0.338	0.635
$W^F \times W^M$	0.233	0.641	0.308
$W^F \times D$	0.413	0.419	0.494
$W^M \times D$	0.278	0.551	0.322

Notes: $\mathcal{A}$ is the overlap coefficient; JS is Jensen–Shannon divergence in bits. D denotes Army/AAF deaths. Computed on county shares of national totals.

I therefore write the investment vector as (W_i^C, W_i^F, W_i^M) and treat military installations as a distinct treatment rather than folding them into durable capital.

One fact about where the war’s productive capacity went belongs here, because it is the mechanism behind everything that follows. Classifying every contract by what it bought

— twenty-one product classes recovered from the description carried on all 190,693 records — and setting each against the county’s 1940 stock of that industry shows that the war’s signature industries were built where they had not been. Forty-three counties held an aircraft establishment in 1940 and 317 received aircraft contracts; 27.9 percent of aircraft dollars went to counties with no prewar aircraft plant at all, 77.2 percent of ammunition dollars, 78.7 percent of small-arms dollars and 89.0 percent of explosives dollars. Where the prewar footprint was already broad the pattern reverses: 5.3 percent of machine-tool dollars and 14.2 percent of textile dollars went to counties without a plant of the kind. Table A20 reports all twenty-one.

Those counties were not empty, and what they did hold is the point. The 278 panel counties that received aircraft contracts without an aircraft plant carried a median 1940 manufacturing share of 0.069 against a national 0.012, and a median thirty-two war-related establishments against four. The war state put its new aircraft and ordnance capacity where there was industry in general but not that industry in particular. The allocation therefore loaded on general industrial capacity rather than on any specific prewar line of production — which is the sharper form of the correlations Table 5 reports, and, as Section 5.9 shows, the reason no instrument built from prewar capacity can identify wartime procurement.

5.3 Alignment

Figure 1 maps the four wartime allocations on 1940 county boundaries; the visual impression is of four surfaces that share little beyond population, and the statistics bear it out — with one exception the table makes plain, contracts and publicly financed plant, which correlate at $\rho = 0.616$ because both followed existing industry. The overlap coefficient between the county distribution of Army/AAF fatal loss and the distribution of contract dollars is

$$\mathcal{A}(p^D, p^{W^C}) = 0.466, \tag{11}$$

with a Jensen–Shannon divergence of 0.338 bits. For durable plant $\mathcal{A} = 0.413$; for military installations $\mathcal{A} = 0.278$. Fewer than half of the two national distributions occupy common ground.

That figure needs a caveat its face value does not carry, and it is better supplied here than left for a referee. Most of the shortfall is not about sacrifice at all. Deaths track population closely — $\mathcal{A}(D, P) = 0.874$ — so the overlap between deaths and contracts is largely the overlap between *people* and contracts, which is 0.498. Decomposing the 0.534 gap between the observed overlap and one: 0.455 of it, 85 percent, is attributable to contracts being concentrated relative to population, a fact established well before this paper (Fishback and

Cullen, 2013). Only 0.079, 15 percent, is attributable to deaths themselves deviating from population.⁸⁹ The evidence for this paper’s claim is therefore not in $\mathcal{A}$, which mostly restates the concentration of war production. It is in what remains once county size is removed, which is what follows.

The coverage asymmetry is stark. Some 98.2 percent of counties recorded at least one Army/AAF death, while 48.5 percent received any major contract: **1,553 counties lost men and received no major war contract at all.** Table A3 lists the twenty largest contract recipients. Concentration is far higher for money than for loss: the twenty counties that lost the most men hold 20.4 percent of the dead, while the twenty largest contract recipients hold 43.8 percent of the contract dollars; at a hundred counties the two figures are 40.7 and 81.3 percent. Loss was distributed almost everywhere; investment was not.

The decisive statistic is what remains once county size is removed. Ranked across counties, the correlation between fatal burden and war investment per resident is -0.026 for contracts, $+0.013$ for industrial plant, and -0.054 for military installations. Three separate allocation mechanisms, administered by different agencies under different statutes, all produce a relationship with sacrifice close to zero. Two of the three are not statistically distinguishable from zero at all. The third is: at $n = 3,073$ the installation correlation of -0.054 has a confidence interval of $[-0.089, -0.018]$, which excludes zero ($p = 0.003$). It does so in the direction that makes the point sharper rather than softer — installations went, if anything, slightly *away* from the counties bearing the heaviest burden — and it accounts for three-tenths of one percent of the variance in installation spending.

The right way to read those three near-zeros is against the criteria the procurement agencies were in fact selecting on, measured the same way, on the same counties, in the same units. Table 5 does that. Per resident, contract dollars correlate with the 1940 manufacturing share at $+0.693$, with the 1940 urban share at $+0.648$, and with 1940 log population at $+0.666$. Publicly financed plant and military installations load on the same characteristics at $+0.23$ to $+0.51$. Against fatal burden all three read -0.026 , $+0.013$ and -0.054 .

This is the paper’s central quantitative statement, and it requires no identifying assumption. The wartime allocation function put weights of roughly 0.2 to 0.7 on prewar industrial capacity,

⁸⁹The two shares are Shapley values, which average each factor’s contribution over both orders in which the two shifts could be applied and so sum to the gap exactly. Shifting one factor at a time and reading off the difference gives 0.408 and 0.032, which together account for only 82 percent of the gap and leave the remaining 18 percent in an unassigned interaction. The Shapley split allocates that term rather than discarding it, and it moves the reading against this paper’s own interest, from 76 to 85 percent of the misalignment being about the concentration of contracts rather than about sacrifice.

⁹Against a Poisson null in which every resident faces the same fatality risk, the observed overlap of 0.466 lies 37 standard deviations below the null mean of 0.498, so the deviation is real. It is simply small. The overlap coefficient is reported because it is the natural summary of distributional correspondence and because the paper’s second question is about correspondence, not because it is where the evidence lies.

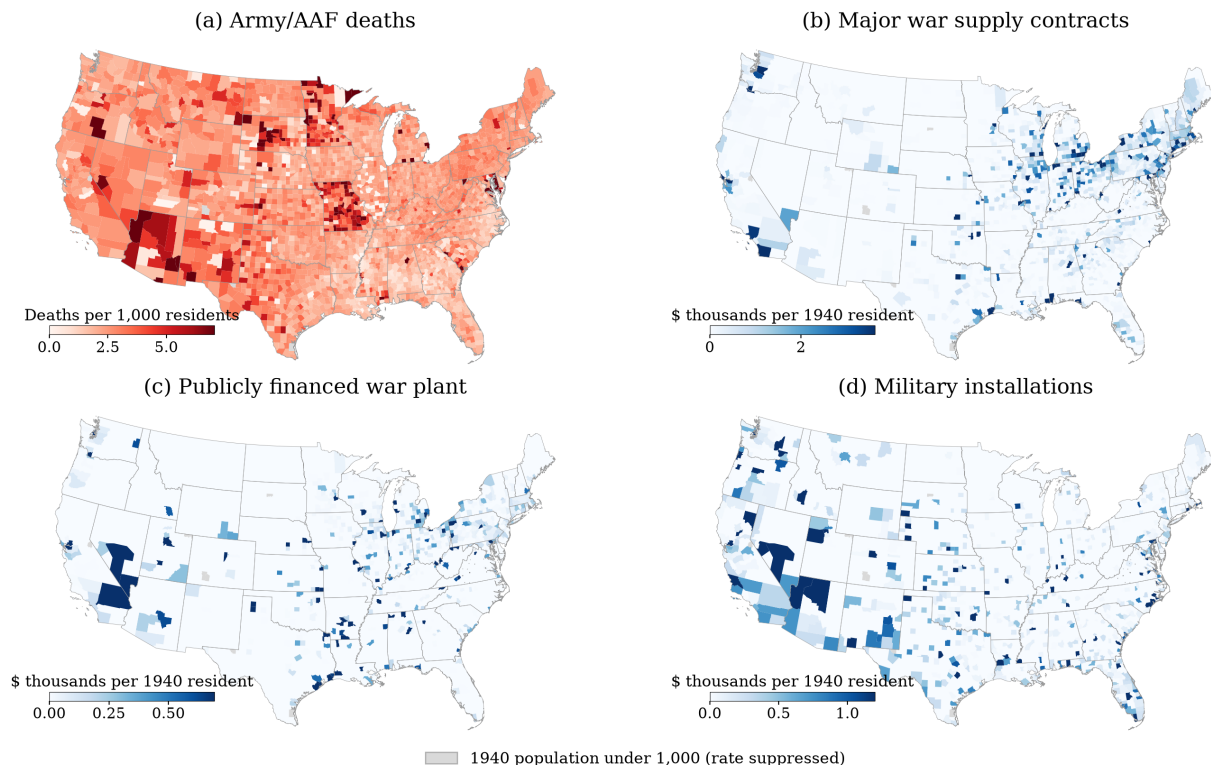


Figure 1: The four wartime allocations on 1940 county boundaries

Notes: Panel (a) plots Army and Army Air Forces deaths per 1,000 1940 residents; panels (b), (c) and (d) plot major war supply contracts, publicly financed industrial plant, and military installations, each in thousands of current dollars per 1940 resident. Human loss is drawn on a red ramp and federal dollars on a blue one, both running pale to dark, so that counties receiving nothing recede rather than dominate the surface; the same two hues form the poles of Figure 2. Note that the four panels carry separate scales and are not directly comparable in magnitude: the comparison of interest is the pattern of coverage, near-universal in panel (a) and sparse in panels (b) through (d). Colour scales are trimmed at the 98th percentile so that a small number of extreme counties do not compress the rest of the distribution.^a Counties with fewer than 1,000 residents in 1940 are shown grey; a handful record impossible rates. Boundaries are the Newberry Atlas of Historical County Boundaries as in effect 1 April 1940, with state outlines drawn in grey; independent cities are merged into their surrounding county. Albers equal-area projection.

^aTrimming affects the display only. Every statistic reported in the text is computed on untrimmed data, with the two exceptions stated explicitly: the 99th-percentile cap on fatal burden before residualizing for Figure 2, and the suppression of counties below a thousand residents from both maps.

urbanization and size, and a weight of essentially zero on human loss. The near-zeros are precise rather than underpowered: at $n = 3,073$ the standard error of a rank correlation is 0.018, so the confidence intervals exclude any association above 0.09 in absolute value, and a true correlation of the magnitude the agencies' own criteria display would have been detected with near certainty.

Table 5: What the wartime allocation loaded on

	Contracts W^C	War plant W^F	Military W^M
	(1)	(2)	(3)
<i>Characteristics the procurement agencies selected on</i>			
1940 manufacturing share	+0.693	+0.445	+0.234
1940 urban share	+0.648	+0.507	+0.398
1940 log population	+0.666	+0.514	+0.328
<i>Military sacrifice</i>			
Fatal burden, per resident	-0.026	+0.013	-0.054
Fatal burden, per enlistee	-0.182	-0.074	-0.193
N , first four rows	3,073	3,073	3,073
N , per-enlistee row	2,976	2,976	2,976

Notes: Spearman rank correlations between each wartime allocation, measured in dollars per 1940 resident, and the row characteristic. Fatal burden per resident is Army/AAF deaths per 1,000 1940 residents; per enlistee is deaths per 1,000 Army/AAF enlistees on the counties passing the validity screen of Section 3. The standard error of a rank correlation at $n = 3,073$ is 0.018.

Conditioning on enlistment sharpens this rather than softening it. Using B_i^A of equation (2)—deaths per thousand men actually enlisted, a national rate of 42.9 with a 90/10 county ratio of 3.58—the correlations turn *negative*, at -0.182 for contracts, -0.074 for plant and -0.193 for military installations against investment per resident (Table 5), and at -0.157 , -0.067 and -0.187 if investment is put on a per-enlistee footing as well.¹⁰ Counties whose soldiers died at higher rates received, if anything, less federal investment per soldier sent. Figure 2 maps the residualized difference between the two exposures, and shows that the counties losing most relative to their investment are not concentrated in any one region — which is what an allocation indifferent to sacrifice, rather than hostile to particular places, should look like.

Table 6 regresses each investment measure on fatal burden and reports three specifications, because the answer depends on which burden measure is used and the difference is instructive.

Without controls, fatal burden predicts none of the three; the only coefficient that reaches even ten percent is a *negative* one on military installations. Adding the prewar vector turns

¹⁰The two sets differ only in the denominator of the investment measure, not in the burden measure. Table 5 reports the per-resident version so that every row of that table shares one set of units.

Table 6: Did fatal burden predict federal war investment?

	Contracts W^C (1)	War plant W^F (2)	Military W^M (3)
B_i , state effects only	-0.0031 (0.0032)	-0.0019 (0.0020)	-0.0027* (0.0015)
B_i , plus prewar controls	0.0106*** (0.0019)	0.0071*** (0.0012)	0.0039** (0.0018)
B_i^A , plus prewar controls	-0.0015 (0.0020)	0.0006 (0.0016)	0.0005 (0.0020)
State fixed effects	Yes	Yes	Yes
N (rows 1–2; row 3)		3,073; 2,976	

Notes: Dependent variables are inverse-hyperbolic-sine transforms of per-capita dollars; each cell is a separate regression. Prewar controls: 1940 log population, urban share, Black population share, manufacturing share, 1930–40 population growth, 1930 illiteracy and farm shares, and indicators for suppressed or imputed controls. All specifications carry state fixed effects. B_i is deaths per 1,000 1940 residents; B_i^A is deaths per 1,000 Army/AAF enlistees and is defined on the 2,976 counties passing the validity screen of Section 3. Standard errors clustered on state in parentheses. *, ** and *** denote significance at the 10, 5 and 1 percent levels.

all three positive and significant. This is worth stating plainly rather than burying, because it is the one place in the paper where a conditional association runs against the headline. What it does not do is overturn it, for two reasons. The first is magnitude: moving a county from the tenth to the ninetieth percentile of its death rate raises the inverse hyperbolic sine of investment per resident by 0.024 for contracts, 0.016 for plant and 0.009 for military installations — on the order of one to three percent of a standard deviation of each, against distributions whose own tenth-to-ninetieth range spans orders of magnitude. The second is that the result does not survive the change of denominator. Under B_i^A , the per-enlistee measure this paper argues is the right one for inference because it holds constant how many men a county sent, no coefficient is distinguishable from zero ($p = 0.45, 0.73, 0.81$). What conditioning on prewar structure recovers is a small positive relationship between a county’s civic death rate and its investment, and that relationship disappears once the number of men exposed is held fixed.

The award dates make it possible to ask whether this is an average concealing a change. Every one of the 190,693 contracts carries an award month, so the allocation can be re-estimated separately for each year of the war, and Table A21 does. It does not change. On the fixed set of 1,490 counties that received a contract at some point, the rank correlation between per-capita contract dollars and the 1940 manufacturing share is 0.497 in 1940 and 0.588 in 1945, moving non-monotonically in between.¹¹ And the coefficient on fatal burden

¹¹The comparable rank correlation computed across all 3,073 counties rises from 0.493 to 0.653, but that movement is mostly coverage: 15.0 percent of counties received a contract in 1940 against 40.1 percent in 1943,

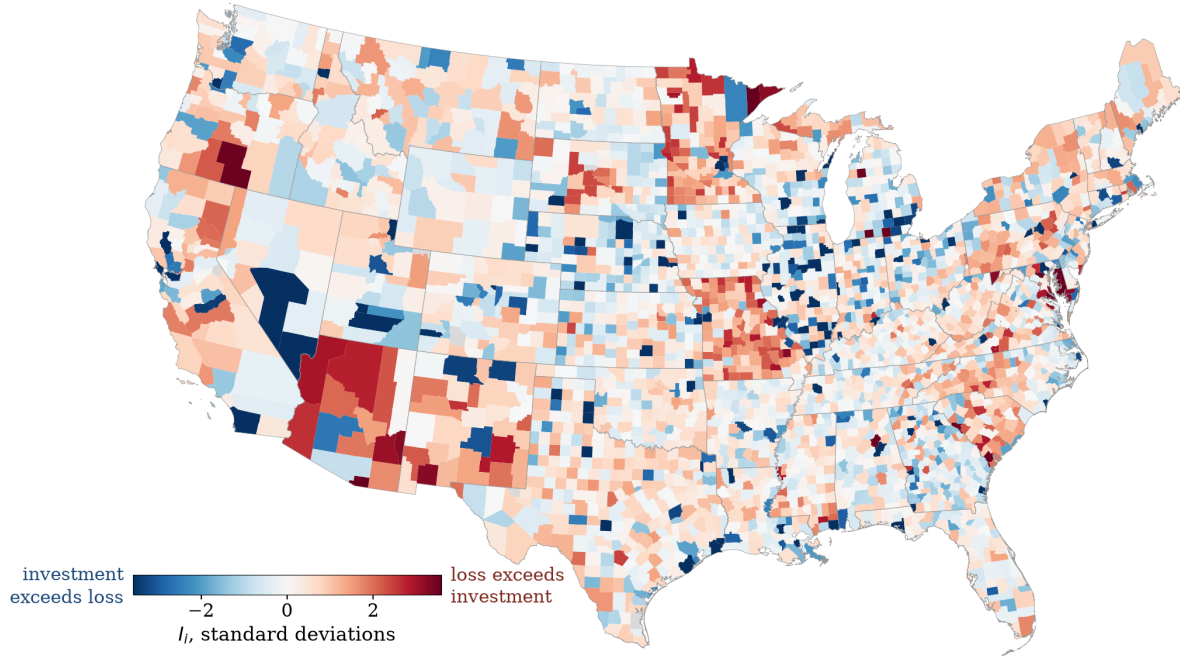


Figure 2: Conditional imbalance I_i between sacrifice and investment

Notes: Both exposures are residualized on 1940 log population, urban share, Black population share, manufacturing share, and 1930–40 population growth, then standardized; I_i is the difference, reported in standard deviations. Red counties lost more than their investment would predict, blue counties the reverse; near-white counties are those where the two exposures correspond. The scale is symmetric about zero and trimmed at the 97th percentile of $|I_i|$. Fatal burden is capped at its 99th percentile before residualizing so that transcription errors in a few very small counties cannot dominate the surface. Boundaries and projection as in Figure 1.

is indistinguishable from zero in every single year under the per-enlistee measure ($p = 0.69, 0.27, 0.38, 0.97, 0.44, 0.83$), on standard errors of 0.0008 to 0.0014 against coefficients never exceeding 0.0012 in absolute value, so the annual zeros are as precise as the pooled one, while under the per-resident measure it is in every year the same small positive that the pooled specification of Table 6 reports. The war state’s indifference to sacrifice is not a property of the average. It held in 1940, when the allocation was small and highly selective, and it still held in 1945.

That B_i^A is not a restatement of B_i^{civic} is worth establishing. The two correlate at 0.628, but they behave oppositely against mobilization intensity: B_i^{civic} is nearly independent of the share of military-age men who served ($\rho = +0.094$) whereas B_i^A is strongly negatively related to it ($\rho = -0.629$). Counties that mobilized more heavily lost a smaller fraction of those they

and a rank correlation over a column that is 85 percent zeros is mechanically attenuated. The fixed-sample figure is the one reported.

sent. A community-burden measure alone conflates sending many men with losing many of those sent; separating them is what equation (2) is for.

5.4 Postwar divergence

Table 7 reports outcome regressions from 1940 through 1970 with prewar controls, state fixed effects, and standard errors clustered on state. Three patterns hold.

First, all three wartime allocations raised county population, and the military coefficient is the largest of the three by 1970 (+0.049 against +0.035 for contracts and +0.020 for plant). Population growth is therefore a poor proxy for the kind of investment received.

Second, and centrally, the two forms of federal capital diverge sharply on industrial structure. On the 1970 manufacturing share, contracts enter at +0.33 and industrial plant at +0.29, while military installations enter at -0.59 . The extensive margin is starker still: merely hosting a war plant is associated with a manufacturing share 1.70 points higher in 1970 ($p = 0.001$), merely hosting a military installation with 3.72 points lower ($p < 0.001$).

That contrast is real but it is not what it first appears, and the levels behind the share say why. All three exposures raise total 1970 employment, and military installations do so as strongly as contracts (+0.042 against +0.041, plant +0.021). On manufacturing employment the ordering is different: contracts enter at +0.060 and plant at +0.027, both at $p < 0.001$, while installations enter at +0.014 and reach only ten percent. The manufacturing share falls in installation counties because their employment grew and their manufacturing did not grow with it, not because manufacturing left. Federal money that arrived as productive plant and federal money that arrived as payroll did not leave the same place behind — but the difference is what each of them built, not that one of them tore something down. Section 5.8 shows the same decomposition inside the matched design.

Tenure is not a mirror of that result and should not be read as one. Military investment enters at -0.34 on the 1970 ownership rate and is the only one of the three distinguishable from zero; industrial plant enters at +0.06 and contracts at -0.05 , neither significant. Installations lowered ownership; plants did nothing measurable to it. That asymmetry holds in the matched design of Section 5.7 as well, and it is the form the contrast actually takes.

Table A5 reports the interactions of equation (10). Three of the four pairs are null. The exception is the 1970 manufacturing share, where $\theta_C = +0.0115$ ($p < 0.001$) and $\theta_F = +0.0036$ ($p = 0.10$): procurement’s association with postwar industrial structure was larger in counties whose soldiers died at higher rates. The magnitude is not trivial — moving across the tenth-to-ninetieth range of B_i^A among counties that received a contract adds roughly 0.6 to a main effect of 0.33 — and it is the one place where fatal burden enters an outcome equation

with force. It does not bear on the allocation result, which is about where investment went rather than what it did, and I report it as an association without a mechanism.

Third, fatal burden predicts essentially nothing on its own. The one coefficient in Table 7 that reaches significance is the 1950 population term, and it is negligible: +0.0009 log points per additional death per thousand residents, so the entire tenth-to-ninetieth range of the county death rate moves 1950 population by under half a percent.

Table 7: Wartime exposure and postwar county outcomes, 1950–1970

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	Log pop. 1950	Log pop. 1970	Mfg. share 1960	Mfg. share 1970	Log income 1950	Log income 1970	Own rate 1970
Contracts W_i^C	0.0118*** (0.0015)	0.0346*** (0.0035)	0.4594*** (0.0833)	0.3325*** (0.1016)	0.0125*** (0.0024)	0.0148*** (0.0016)	-0.0483 (0.0902)
War plant W_i^F	0.0103*** (0.0025)	0.0197*** (0.0046)	0.4165*** (0.0752)	0.2851*** (0.0829)	0.0087** (0.0034)	0.0101*** (0.0025)	0.0562 (0.0861)
Military W_i^M	0.0175*** (0.0017)	0.0493*** (0.0047)	-0.4558*** (0.0835)	-0.5945*** (0.1108)	0.0121*** (0.0027)	0.0086*** (0.0017)	-0.3360*** (0.0547)
Fatal burden B_i	0.0009*** (0.0002)	-0.0023 (0.0026)	-0.0968 (0.0650)	-0.0798 (0.0961)	-0.0001 (0.0023)	-0.0013 (0.0010)	-0.1335 (0.1164)
Prewar controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes
State fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes
N	3,072	3,065	3,071	3,066	3,010	3,066	3,066
R^2	0.984	0.929	0.745	0.697	0.774	0.641	0.472

Notes: All specifications include the full prewar control set and state fixed effects. Standard errors clustered on state in parentheses. Income outcomes are logged. Coefficients are conditional associations, not causal effects; see the pre-trend evidence in Table A4. *, ** and *** denote significance at the 10, 5 and 1 percent levels.

Across population, manufacturing, income, and homeownership, and under four alternative denominators—residents, adult males, men in the labor force, and battle deaths only—no specification yields a coefficient distinguishable from zero at conventional levels (Table A6). What a county gave the war did not shape what it kept from it.

5.5 Whose sons, and in which branch

The enlistment file records race, education, birth year and enlistment source for every soldier, and the Army General Classification Test score for 5.4 percent of them, which makes it possible to ask whether county differences in fatality rates reflect differences in risk or differences in who was sent. The test-score term should be read with that coverage in mind: a county’s mean rests on roughly one enlistee in twenty, and 139 counties carry no scored record at all and take the sample median. Restricting to the 2,966 counties with at least a hundred enlistees — covering 7.17 million enlistees and 288,059 deaths — and regressing the county fatality rate on the composition of its soldiers gives $R^2 = 0.177$.

The coefficients are informative in themselves. A larger volunteer share raises the county fatality rate sharply, a larger share of Black enlistees lowers it, and higher average education and test scores raise it. All four run through assignment to branch: volunteers and high-scoring men were channeled toward the Army Air Forces and the airborne divisions, while Black soldiers were concentrated in service and supply units under the segregated manpower policy of the period, a pattern consistent with the occupational evidence in Ferrara (2022). A county’s fatality rate was therefore not only a function of how many men it sent but of which parts of the Army they entered.

Composition accounts for 9.3 percent of the cross-county standard deviation in fatality rates, so the great majority of the variation survives adjustment. Residualizing the rate on composition does not weaken the paper’s central result; it strengthens it. The rank correlation between composition-adjusted fatality risk and per-capita investment is -0.182 for contracts, -0.119 for industrial plant, and -0.083 for military installations. For contracts and plant the adjustment makes the correlation more negative, not less (-0.170 and -0.070 before it); for military installations it moves the other way, from -0.178 to -0.083 .¹² All three remain negative under either treatment. Counties whose men faced higher risk conditional on who they sent received no more federal investment, and by these estimates slightly less.

5.6 What the estimates can and cannot support

Two results discipline the interpretation. Spatial standard errors computed with a Conley kernel at 200 kilometers leave every sign and significance level in place, so the findings are not an artifact of treating neighboring counties as independent (Table A7)¹³; and dropping the twenty largest contract recipients changes no sign or significance and moves no coefficient by more than five percent of its own value (Table A8), so they are not driven by a handful of arsenal counties.

A source limitation should be recorded. The Army/AAF honor list is not complete at county level: matched to the 1940 census, 56 of 3,073 counties carry no recorded death, and they are concentrated in Illinois, where 80 of 102 counties appear and the implied death

¹²The unadjusted figures here differ slightly from the per-enlistee row of Table 5 because this exercise runs on the 2,966 counties with at least a hundred enlistees and uses the fatality rate built from the DS2 decode, which is the universe the individual composition variables are defined on.

¹³Two hundred kilometers is roughly the radius within which a county’s labor market and its wartime procurement neighbors are likely to overlap. Six of the eight spatial standard errors are tighter than the state-clustered ones, though one of the six only in the fifth decimal; the two that are not are contracts on 1970 population, where the Conley error is 0.0042 against 0.0034 and the t statistic is still 8.25, and war plant on the 1970 manufacturing share, at 0.0873 against 0.0829 and $t = 3.26$. The state-clustered errors are reported throughout because they are the more conservative of the two in six of eight cases and the exceptions change nothing.

rate is 1.90 per thousand residents against a national 2.26. Nebraska is next with seven. Independent cities are split inconsistently between city and surrounding county in Maryland, Missouri, and Virginia. Excluding Illinois entirely leaves every sign and every significance level in Table 7 intact. The largest movement is $+0.018$, on the contract coefficient for the 1960 manufacturing share (0.459 to 0.477), which is about a fifth of that coefficient’s standard error; $\mathcal{A}$ shifts from 0.466 to 0.454. The state matters enough to be worth reporting and not enough to carry a result. The cross-state variation that remains is substantially real rather than artifactual: New Mexico’s rate of 3.61 per thousand, the highest of any state, reflects the destruction of the 200th Coast Artillery on Bataan.¹⁴

Against this, a placebo test is partly unfavorable. Wartime exposure was assigned between 1940 and 1945 and so should not predict earlier outcomes, yet contracts and military installations both predict 1930–40 population growth, and both predict the prewar manufacturing share — contracts positively and installations negatively (Table A4). Federal investment went disproportionately to counties already growing. Contract dollars went to counties already industrial; installations went away from them, which is the same divergence Section 5.2 reports and not a second confound.

Carrying the same test back one further decade bounds what that means. Against 1920–30 population growth — before the Depression as well as before the war — no exposure predicts anything: contracts enter at $+0.0004$ ($p = 0.85$), plant at $+0.0044$ ($p = 0.26$), installations at $+0.0008$ ($p = 0.85$), and fatal burden at -0.0004 ($p = 0.81$), on the 3,041 counties whose 1920 population can be matched.¹⁵ The counties that later received war investment were not on a different trajectory in the 1920s. What Table A4 records is a Depression-decade divergence, which is a narrower and more tractable problem than a long-running one, though it remains a reason to read Table 7 as conditional association. Fatal burden runs the other way: casualty-heavy counties were already losing population before the war. The estimates in Table 7 are therefore reported as conditional associations on a rich prewar control set, not as causal effects. The industrial-plant pre-trend is the mildest of the three, which is a point in favor of the publicly financed plant design of Garin and Rothbaum (2025); Section 5.7 builds a quasi-experimental treatment on G_i and reports it.

¹⁴The 200th Coast Artillery (Anti-Aircraft) was a New Mexico National Guard regiment federalized in 1941 and sent to the Philippines. Roughly half its men died in captivity after the surrender of Bataan, and the regiment drew disproportionately from a small number of New Mexico counties. This is the clearest illustration in the data of the paper’s point about branch and unit assignment: a county’s fatality rate depended on which formation its men entered, not only on how many it sent.

¹⁵From Haines part 24. The match totals 105.7 million residents against a 1920 continental census of 105.7 million; the thirty-one counties that fail to match were created after 1920, and one further county lacks a usable 1920 figure. Together they hold 0.14 percent of 1940 population.

One further hypothesis was tested and rejected. If loss and investment diverged, a natural conjecture is that casualty-heavy counties received bases rather than plants—payroll rather than productive capital. The raw pattern is suggestive: among counties receiving one type only, 34.2 percent of the lowest-burden quartile received a plant against 54.9 percent of the highest. The association does not survive prewar controls and state fixed effects: the coefficient on fatal burden falls to 0.0002 with a standard error of 0.0029 and $p = 0.95$, and it is reported here as a null. The divergence is not that sacrifice bought the wrong kind of investment. It is that sacrifice bought no particular kind at all.

5.7 A matched comparison of large public plants

The instrument fails because prewar industrial capacity predicts both treatment and outcome. A design that does not require an instrument can still succeed if it compares treated counties to counties that look like them on exactly that dimension. This is the strategy of Garin and Rothbaum (2025), and I implement it here.

Treatment. A county is treated if it received at least one plant that was publicly financed (private share of structure spending at most five percent), newly built (structures above forty percent of plant cost), and large (cost of at least \$10 million). The facilities report marks new construction with an explicit **New Plant** designation printed beneath a facility’s location; combined with the structure-share rule this yields 226 qualifying plants worth \$8.61 billion, of which 214 resolve to a county. Those 214 are worth \$8.08 billion, are spread across 164 counties after the independent-city merge, and carry 93.9 percent of the qualifying dollars. Measured against the base Garin and Rothbaum (2025) report on — spending on all newly built wartime plants, publicly and privately financed, which the facilities volume puts at \$11.94 billion — the 226 capture 72.1 percent, against roughly 70 percent for their 353-plant definition. The two selections therefore cover the same ground; the count differs because the unit of record here is a cost line while theirs is a deduplicated plant identifier.

Sample. The hundred largest manufacturing counties of 1940 are dropped. These are the places whose selection into treatment is least comparable to anything else, and retaining them means estimating a contrast between Detroit and rural Georgia.¹⁶ After this restriction the sample is 2,973 counties, of which 118 are treated.

Matching. A propensity score is estimated on the prewar vector — 1940 log population, urban share, Black population share, manufacturing share, 1930–40 population growth, 1930 illiteracy and farm shares, and latitude and longitude, with squared terms — and treated

¹⁶The restriction also removes the region of the propensity distribution where common support is thinnest. It is imposed on the estimation sample throughout rather than varied as a robustness check, so no result in the paper rests on a comparison involving the largest prewar manufacturing centers.

counties are matched to at most three controls within a caliper of 0.2 standard deviations, on common support.¹⁷ 115 of 118 treated counties match to distinct control counties.

Balance. Table A9 reports standardized mean differences. Before matching, seven of the nine covariates exceed the conventional 0.10 threshold, and log population differs by 1.45 standard deviations. After matching, **only two still do**. The two that do are latitude (−0.124), a geographic coordinate rather than a measure of economic structure, and the 1940 Black population share (0.107), which sits just past the threshold and enters the outcome regressions as a control in any case. On prewar population, urbanization, industrial structure and growth — the dimensions on which selection into treatment actually operated — the treated and control groups are balanced to within 0.10 standard deviations.

Estimates. Table 8 reports the results. Receiving a large new publicly financed plant is associated with a 1970 population 23.3 percent higher, a manufacturing share 3.34 points higher, and family income 7.3 percent higher, all relative to matched controls and all significant at the one percent level. Homeownership is the exception and is reported as a null: the point estimate is +0.94 points with a standard error of 0.61, which does not reach significance at any conventional level. Inverse-probability weighting on common support gives the same signs throughout and larger magnitudes for population, manufacturing and income.

Table 8: Effect of a large new publicly financed wartime plant, matched comparison

Outcome	Raw difference	Matched	(SE)	IPW	<i>N</i>
log population 1970	1.3970	0.2092***	(0.0605)	0.3041	364
manufacturing share 1970	3.9528	3.3378***	(0.8458)	3.6874	364
log family income 1970	0.2201	0.0704***	(0.0194)	0.1115	364
homeownership 1970	-1.6707	0.9416	(0.6117)	0.5376	364
log family income 1950	0.2865	0.0647***	(0.0189)	0.1011	364
manufacturing share 1960	6.4296	3.2824***	(0.8310)	4.1448	364
log total employment 1970	1.4425	0.2228***	(0.0646)	0.3044	364
log manufacturing employment 1970	1.7126	0.3878***	(0.0854)	0.5126	364

Notes: 115 treated counties matched to 249 distinct controls within a 0.2 standard-deviation caliper on the propensity score, excluding the 100 largest manufacturing counties of 1940. Matching is three-to-one with replacement; *N* is the number of distinct counties on the matched sample. Matched estimates are weighted least squares on the matched sample with the full prewar covariate vector; IPW is inverse-probability weighting on common support. Standard errors clustered on state. Balance in Table A9; placebo tests in Table A10; robustness in Table A11. *, ** and *** denote significance at the 10, 5 and 1 percent levels.

Placebo. Table A10 applies the same estimator to outcomes determined before the plants were sited. On the two that describe prewar economic structure treatment predicts

¹⁷The caliper follows the convention of one-fifth of a standard deviation of the propensity score. Table A11 varies it from 0.1 to 0.5 and varies the number of matched controls from one to five; the 1970 population coefficient stays within 0.193 to 0.211 across all five.

nothing: 1930–40 population growth ($p = 0.66$) and the 1940 manufacturing share ($p = 0.81$) are indistinguishable between treated and matched control counties. This is the test the instrument failed, and on those dimensions the matched design passes it.

The third is weaker and deserves saying plainly. Treated counties had a 1940 homeownership rate 1.27 points above their matched controls, at $p = 0.11$ — short of significance at five percent, but not comfortably so, and larger in magnitude than the +0.94 the same design estimates for 1970. A pre-treatment coefficient that exceeds the post-treatment one is not evidence of an effect; it is a reason to treat tenure as the dimension this design does not identify. That is the conclusion the paper reaches below on independent grounds, and the placebo is why it is reached rather than assumed.

What survives the hardest comparison. Table A11 varies the matching parameters and the sample. Results are stable across one-, three- and five-nearest-neighbor matching, across calipers of 0.1 to 0.5 standard deviations, and across restrictions of the sample to counties above ten thousand residents or with any 1940 manufacturing employment; the 1970 population coefficient stays within 0.193 and 0.217 across all seven. The demanding test is the last panel, which compares treated counties only to their own untreated neighbors. The control pool is restricted to the 475 untreated counties that directly border a treated one — places sharing a labor market, a climate, and a state — and 111 of the 118 treated counties find a match inside it. There the population effect (+0.180, $p = 0.002$), the manufacturing effect (+1.63, $p = 0.022$) and the income effect (+0.045, $p = 0.012$) all survive, at magnitudes between roughly half and the whole of the matched-control estimate. Homeownership is again indistinguishable from zero (+0.40, $p = 0.48$).

One quantity should be attached to that claim before it is made. Both designs are selection on observables, and Section 5.9 argues at length that prewar industrial capacity reaches 1970 outcomes by routes that do not pass through wartime production. Oster (2019) gives the natural bound: how strong would selection on unobservables have to be, relative to selection on the prewar vector already in the model, to drive each estimate to zero? Table A12 reports it, and the answer separates the two designs.

The plant estimates are the more fragile. At Oster’s own $R_{\max}^2 = 1.3R^2$, δ is 3.50 for the manufacturing share and 1.51 for family income, both above the conventional threshold of one — but 0.98 for population and 0.86 for manufacturing employment, both just below it. Under the stricter $R_{\max}^2 = 1$ only the manufacturing share survives. The table shows why: the raw plant-county gap in 1970 log population is 1.78, and the prewar vector alone takes it to 0.23. When the observables already account for that much of a raw association, Oster’s arithmetic says unobservables of comparable strength would account for the rest. This is the

same fact Section 5.9 argues from a different direction, and it is a real qualification rather than a technicality.

The military estimates are the more robust, and two of them robust in the stronger sense. For population (-19.9) and family income (-26.4) δ is negative, because adding the prewar vector moves those coefficients *away* from zero rather than toward it: observed selection runs against the estimated effect, so unobserved selection would have to be correlated in the opposite direction to overturn it. For the manufacturing share and for homeownership δ is 4.70 and 23.25, and remains above one at $R_{\max}^2 = 1$ (1.66 and 4.32).

I therefore claim causal identification unevenly. The plant effect on the manufacturing share is robust on this test at either $R_{\max}^2$; the effect on family income is robust at Oster’s and not at the stricter bound; the effects on population and on manufacturing employment sit at the threshold, and I report them as estimates that a referee is entitled to discount. Large publicly financed wartime plants raised the population, the manufacturing employment, and the family income of the counties that received them, and those effects persist to 1970 and survive comparison against immediate neighbors — but only the industrial composition among them is insulated from a confounder as strong as prewar industrial capacity itself. They did not measurably change the rate at which those counties’ residents owned their homes, against either comparison group.

This matters for the paper’s central question. The counties that received durable public capital did acquire lasting economic advantages — and Section 5 shows that which counties those were had essentially nothing to do with which counties bore the war’s human cost.

5.8 The same design applied to military installations

Garin and Rothbaum (2025) establish the long-run effect of large publicly financed industrial plants. No comparable estimate exists for the war’s other great federal capital allocation. Section 5 showed that industrial plant and military installations occupy weakly related geographies and are associated with opposite postwar structures. That association is what this section tests.

Treatment is membership in the top decile of per-capita military facility spending in the 1947 County Data Book, taken among the counties that received any such spending and within the estimation sample. Fifty-six counties clear the threshold; two of them also received a large public plant and are dropped so the two treatments do not contaminate one another, leaving the 54 the design actually studies. Those 54 hold \$2.01 billion, 25 percent of the military facility dollars in the sample they are drawn from and 20 percent of the national total. Matching proceeds exactly as in Section 5.7: 52 of the 54 match to 152 controls, and the

treated counties are spread across 27 states with at most four in any one, imbalance falls from eight of nine covariates to one, and all three placebo tests are clean — 1930–40 population growth ($p = 0.66$), the 1940 manufacturing share ($p = 0.63$), and the 1940 home-ownership rate (+0.53 points, $p = 0.70$), which is the pre-treatment outcome the design’s headline result most needs (Table A13).

Table 9 sets the two treatments side by side, and the contrast is the paper’s sharpest result. A military installation raised county population by 1970 far more than a war plant did — 55 log points against 21 — while *reducing* the manufacturing share by 4.72 points where the plant raised it by 3.34. Family income rose under both. On homeownership the two designs do not mirror each other: the installation effect is -7.49 points and the plant effect is a null, so the tenure result belongs to installations alone.

The manufacturing rows carry a distinction the share alone cannot make. A share falls either because its numerator shrinks or because its denominator grows, and those are different histories. The employment levels separate them, and the answer is the second. Total 1970 employment rose under both treatments, by 41 log points for an installation and 22 for a plant. Manufacturing employment rose by 39 log points under a plant ($p < 0.001$) and by 22 under an installation, where it is imprecise rather than zero ($p = 0.10$). Installations did not deindustrialize the counties that received them; they grew everything else around a manufacturing base that did not move, and the falling share is that arithmetic. The plant is the opposite case: its manufacturing employment grew faster than its employment as a whole, which is why the share rose. So the two treatments do not have opposite effects on manufacturing. One of them has an effect on manufacturing and the other has an effect on everything else — which is the sharper statement, and the one the county panel can actually support.

Two forms of federal capital, arriving in the same years under the same war, therefore left different places behind. The installation counties grew faster and their residents earned more, but they grew in everything except manufacturing and they did so with markedly lower rates of ownership. The plant counties grew less, industrialized, and left tenure where they found it.¹⁸ In 1943 both looked like the same thing: the federal government had arrived, and the community was contributing to the war.

One limit on the tenure result must be stated, because the census counts occupied dwelling units and a large installation puts a standing population of service members and their families into rented quarters on and around it. Some part of the -7.49 points is therefore composition

¹⁸The plant coefficient on 1970 homeownership is +0.94 points against matched controls ($p = 0.12$), +0.40 against untreated neighbors ($p = 0.48$), and indistinguishable from zero in the continuous specification of Table 7. The installation coefficient is -7.49 points and survives every specification tried. The asymmetry is stated in the conclusion.

Table 9: Two forms of federal capital, matched designs

Outcome, 1970	Large public plant		Military installation	
	Effect	(SE)	Effect	(SE)
Log population 1970	0.2092***	(0.0605)	0.5509***	(0.0792)
Log total employment 1970	0.2228***	(0.0646)	0.4141***	(0.0854)
Log manufacturing employment 1970	0.3878***	(0.0854)	0.2151	(0.1318)
Manufacturing share 1970	3.3378***	(0.8458)	-4.7201***	(1.3957)
Log family income 1970	0.0704***	(0.0194)	0.1051***	(0.0255)
Homeownership 1970	0.9416	(0.6117)	-7.4890***	(1.1235)
Treated counties	115		52	
Covariates imbalanced	2 of 9		1 of 9	

Notes: Each column pair is a separate matched design estimated on its own propensity score, excluding the 100 largest manufacturing counties of 1940 and, for the military design, counties receiving both treatments. Weighted least squares on the matched sample with the full prewar covariate vector; standard errors clustered on state, 44 state clusters for the plant design and 40 for the military design, of which 38 and 27 respectively carry a treated county. Because 27 is thin for cluster-robust inference, the military estimates are reported alongside the randomization inference of Table A14, which assumes nothing about the cluster structure. The first three rows are in logs; manufacturing employment is the 1970 manufacturing share applied to 1970 total employment, and the two employment rows decompose the share row beneath them rather than adding a finding to it. Placebo tests in Tables A10 and A13.

rather than a change in what the civilian population could acquire, and a county panel cannot separate the two. Three facts describe the gap without discriminating between the two readings, and it is worth being explicit that they do not. It is not pre-existing: treated and matched control counties are indistinguishable on the 1940 ownership rate (+0.53 points, $p = 0.70$). It is not confined to the mobilization years: it is -5.57 points already in 1950, -6.35 in 1960 and -7.49 in 1970, widening across a quarter-century in which the national ownership rate rose steeply. And it sits beside a 1970 median house value 15 percent *higher* than controls'. Each of the three is equally what a standing garrison predicts and what a durable change in the civilian housing market predicts, so none of them separates the two, and a county panel of occupied dwelling units cannot.

What the estimate does establish is the narrower thing. Counties that received the war's payroll capital carried a permanently more rental-tenured housing stock through 1970, and by then most of these installations had been open and manned for a quarter-century, through Korea and the Cold War buildup. That is a fact about what those places were, not a measurement of what their civilians could acquire. Separating the two would need the tenure of civilian households specifically, or the treated counties split by whether the installation survived to 1970, and this panel supports neither.

The military result is not an artifact of where the treatment threshold is drawn. Table A14 re-estimates it at the top five, ten, twenty, and thirty percent of per-capita military spending and at an absolute \$10 million cutoff. Every definition gives the same signs at $p < 0.01$, and the magnitudes fall as the threshold is relaxed: the homeownership effect declines monotonically from -9.25 points at the top five percent to -4.29 at the absolute cutoff, and population likewise from 0.87 to 0.37 log points. The manufacturing effect falls from -9.09 to roughly -4.3 and is then flat rather than monotone, moving from -4.31 at the top twenty percent to -4.61 at the top thirty. Two of the three outcomes therefore show the dose-response pattern one expects of a real effect rather than of a threshold artifact, and the third shows a large drop followed by a plateau. Because the treated group is small, I also report randomization inference: permuting treatment 500 times gives permutation p -values of¹⁹ 0.000 for population, 0.004 for the manufacturing share, and 0.000 for homeownership.

The GI Bill. Between treatment and outcome sits the largest veteran-specific federal intervention in American history, and it operates on two of the four outcomes estimated here. Fetter (2013) attributes 7.4 percent of the 1940–60 national rise in home ownership to the mortgage guarantees of the World War II and Korean War GI Bills, and 25 percent of the rise for affected cohorts, working primarily by moving purchase earlier in the life cycle; Bound and Turner (2002) find moderate gains in postsecondary attainment for returning veterans. A referee should ask whether the 1970 home-ownership and income estimates are partly GI Bill estimates.

The benefits flowed to veterans, so the channel is open only if treatment predicts a county’s veteran population. It does not, for the plant design. Regressing enlistees per thousand 1940 residents on plant treatment with the full prewar control vector gives -3.63 ($p = 0.46$), and deaths per thousand enlistees gives $+2.56$ ($p = 0.79$): treated and matched control counties sent men at indistinguishable rates and lost them at indistinguishable rates. Whatever the GI Bill did to home ownership in these counties, it did in both arms.

For the military design the answer is different, and it runs against the finding rather than toward it. Installation counties sent 10.2 more enlistees per thousand residents than their controls ($p = 0.033$), while the fatality rate per enlistee is again indistinguishable ($p = 0.78$). More veterans returning to a county with GI Bill loan guarantees in hand should, on Fetter’s estimates, have *raised* its home-ownership rate. The measured effect of a top-decile installation on 1970 home ownership is -7.49 points. The GI Bill channel therefore biases

¹⁹Treatment is reassigned at random among counties on common support, holding the number of treated units fixed, and the full matching and estimation pipeline is re-run on each draw. The reported p -value is the share of draws producing an absolute effect at least as large as the observed one, so it requires no distributional assumption about the 52 treated units.

that estimate toward zero, and the structural divergence between the two forms of federal capital is if anything understated.

Two further checks bear on the paper’s central claim rather than on this design. Of the 2,973 counties in the sample, 118 received a large public plant and 56 a top-decile installation, and only two received both — against 2.2 expected if the two allocations were statistically independent (Fisher exact $p = 1.00$). The war state’s two capital programs did not merely fail to coincide with sacrifice; they did not coincide with each other. And of the eighteen treatment-effect estimates the two designs report, fifteen survive a Bonferroni correction for the full set of tests at $0.05/18 = 0.0028$; the three that do not are the plant effect on homeownership ($p = 0.124$), the military effect on manufacturing employment ($p = 0.103$) and the military effect on the 1960 manufacturing share ($p = 0.075$).²⁰

This is why the paper insists that wartime investment is a vector rather than a scalar, and it is what gives the alignment result its force. Whether a county inherited an industrial base or a service economy from the war turned on which kind of federal capital it received — and Section 5 shows that which counties received which had essentially nothing to do with which counties bore the war’s human cost.

5.9 An instrument, and why it does not work

The estimates above are conditional associations. The natural route to a causal claim is the 1938 Industrial Mobilization Plan. In that year the Joint Army and Navy Munitions Board surveyed American industry and allocated named establishments to specific procurement branches against the possibility of war, publishing the result as a directory of facilities allocated and reserved. The allocation was made before the war, so it cannot be caused by wartime spending; and Fishback and Cullen (2013) and Jaworski (2017) both use it as a measure of prewar military-production capacity. The same source records prewar Army and Navy establishments and bases, which offers a separate instrument for W^M . I recover 8,810 allocated facilities across 833 counties and 529 prewar military sites across 133.

²⁰The family is every treatment-effect estimate the two matched designs report: population, the manufacturing share, family income and home ownership in 1970 under both treatments, total and manufacturing employment in 1970 under both, 1950 family income and the 1960 manufacturing share under the plant design, and the 1960 manufacturing share, 1950 and 1960 home ownership and the 1970 median house value under the military one. An earlier version excluded the employment-level rows on the ground that they decompose the manufacturing-share estimate; that exclusion cannot stand, because the paper treats manufacturing employment as a finding in its own right and reports the share as the quantity it does *not* claim. Widening the family costs nothing the paper relies on: the estimate it adds to the failing set, the military effect on manufacturing employment, is one the paper already reports as a null. The alignment statistics are descriptive and require no correction, and the placebo and robustness tests are diagnostics.

The F statistics are not the problem. Table A15 reports first-stage F statistics of 23.5 for contracts on allocated facilities, 63.6 for war plant on the industrial allocation, and 18.9 for military installations on prewar bases, all conditional on the prewar control set and state fixed effects.

Validity is the problem, and the diagnostics say so in three ways.

First, the three-endogenous system is weakly identified despite those F statistics. The partial R^2 of the three excluded instruments, after the prewar controls and state effects are absorbed, is 0.030 for contracts and 0.013 for military installations: the instruments explain under three percent of the residual variation in two of the three treatments. Only war plant, at 0.069, clears the five percent this paper elsewhere treats as a minimum, and it is the treatment whose instrument fails the exclusion test most badly. The resulting 2SLS coefficients are implausibly large and reverse the OLS signs.

Second, and more tellingly, the sign reversal is common to all three instruments. Instrumenting war plant alone gives a 1970 homeownership coefficient of -5.29 ; instrumenting contracts alone gives -8.95 ; instrumenting military installations alone gives -7.44 (Table A16). Three different treatments, instrumented by three different prewar measures, do not all have large negative effects of the same order on the same outcome — one on which two of the three have no OLS effect at all. What they share is the instrument’s content.

Third, the placebo test makes that content explicit. Table A17 regresses *pre-war* outcomes on the instruments with the same controls. The industrial allocation is strongly related to the 1940 manufacturing share ($p < 0.001$) and the 1940 urban share ($p < 0.001$); so are the prewar bases. It is unrelated to 1930–40 population growth, so the instrument is not picking up a prewar trend. It is picking up a prewar *level*: the Munitions Board allocated plants that were already industrially capable, in places that were already urban and already manufacturing. Those characteristics have many routes to 1970 outcomes that do not pass through wartime production, and the exclusion restriction fails.

Is the failure the construction or the source? A referee is entitled to ask whether a different construction of the instrument would survive. Table A18 answers systematically, sweeping nine constructions from the broadest — every establishment allocated to any procurement branch — to the narrowest, the aeronautical and optical branches alone, which allocate against a specific technical capability rather than against general manufacturing capacity. For each it reports the first-stage F , the partial R^2 after controls and state effects, and placebo regressions on four pre-war outcomes.

None of the nine is usable. The pattern is a trade-off rather than a sequence of unlucky draws. The narrow-branch instruments do what one would hope on two of the four placebos: aeronautical allocations are unrelated to the 1940 manufacturing share ($p = 0.16$)

and to the 1940 urban share ($p = 0.21$), so they are not simply a restatement of prewar industrial structure. But their partial R^2 is 0.003, and they still predict 1940 log population at $p < 0.001$ — the Munitions Board put aircraft work where there were people to do it. The one construction with adequate residual variation, an indicator for any allocated facility at a partial R^2 of 0.055, fails all three level placebos at $p < 0.001$. Every instrument narrow enough to escape prewar industrial structure is too weak to identify anything; every instrument strong enough to identify carries that structure with it.

Is the failure the Munitions Board, or the war? The nine constructions above read one 1938 document. A shift-share assembled from different sources fails the same way. Interacting each county’s share of the national 1940 stock of an industry with national contract dollars in the matching product class — twenty-one classes recovered from the description carried on every contract record — gives a materially stronger instrument, $F = 45.1$ against 19.7 and a partial R^2 of 0.039 against 0.014, which still fails all three level placebos at $p < 0.001$ (Table A19).²¹ Conditioning in addition on the total 1940 stock of war-related establishments buys the manufacturing-share placebo, at $p = 0.87$, and costs the relevance that made the instrument worth having: F falls to 11.3 and the partial R^2 to 0.013, while the urban share and log population still reject at $p < 0.001$. That is the same relevance-against-exclusion frontier the nine constructions trace, reached from a different archive by a different method, and it is what makes this a statement about the setting rather than about the Munitions Board.

Why it fails is Section 5.2’s fact read backwards. A shift-share is a weighted sum of share instruments, so its identification rests on whichever shares carry the weight (Goldsmith-Pinkham et al., 2020), and here that is textiles at a partial R^2 of 0.037 and machine tools at 0.032, while aircraft, at \$51.9 billion the largest class in the file, contributes 0.0001 and ammunition at \$14.4 billion contributes 0.0000 — both on all records, against the geocoded totals Table A20 prints. A prewar-capacity instrument reaches only the procurement that ran through plants which already existed — and that, as Table A20 showed, is the minority of the war’s signature spending and the majority of what prewar industrial structure explains.

I therefore do not report instrumented estimates as causal. This is a substantive conclusion about the setting rather than a technical failure of one specification. The placement of wartime industrial investment was not a lottery, and the best available prewar predictor of

²¹The county shares are 1940 establishment counts by industry from the Jaworski (2017) archive for the eighteen classes it carries, and 1930 county employment for machinery, electrical equipment and food, which have no 1940 establishment counterpart there; Table A20 marks the three. Iron and steel accordingly enters twice, as 1940 establishments for the primary-metals class and as 1930 employment for machinery, so those two rows are not independent. Construction, the leave-out variants and the product classifier are in the replication package.

that placement is prewar industrial structure itself — which is precisely the confounder any causal claim must overcome. The credible design must instead come from comparing treated counties to untreated counties that resemble them on exactly that dimension, which is what Sections 5.7 and 5.8 do.

6 Shared Sacrifice and the Political Compression of Geography

The second question stated in Section 1 asks what the measured geography does to the wartime claim of shared sacrifice. It is not a question the regressions can answer, but it is one they now constrain.

Roosevelt’s April 1942 fireside chat did not describe sacrifice as a singular military act. He placed higher taxes, wage and price limits, rationing, bond purchases, war work, and military service inside a single moral frame, an “economy and equality of sacrifice” (Roosevelt, 1942). The Office of War Information elaborated the same grammar visually, teaching civilians to read consumption restraint and factory labor as continuous with battlefield service (Winkler, 1978; Sparrow, 2011); the surviving poster series make the equivalence explicit, addressing the war worker and the bond buyer in the vocabulary reserved for the soldier (NARA, n.d.; NMAH, n.d.). That civilian contribution was materially real as well as rhetorical: war-bond purchases displaced roughly \$70 of bank deposits per \$100 sold (Brunet et al., 2025a,b). The operation is one of commensuration: the soldier, the riveter, the taxpayer, and the rationing household are made members of one moral community by being made contributors of one commensurable thing.

Commensuration of that kind requires a scale, and a scale requires that the quantities compared be of the same type. The measured geography shows they were not, and shows it in a specific way. Had the finding been that investment tracked loss, the rhetoric would read as description. Had it been that investment ran *inversely* to loss, the rhetoric would read as inversion—a claim of equality covering a systematic transfer away from the places that bore most. The actual result is neither. Across three separate allocation mechanisms the rank correlation with sacrifice is approximately zero. The war state did not reward sacrifice, and it did not punish it. It was, with respect to sacrifice, *indifferent* — a word for the pattern in the allocation, not for a disposition anyone held. Nothing here distinguishes a state that considered sacrifice and set it aside from three procurement processes optimizing over criteria that had nothing to do with one another, and the second is much the more likely.

Indifference is the harder case for the rhetoric, and the more interesting one. A transfer away from high-casualty places would have been a betrayal of the claim in its own terms, and answerable in those terms. Orthogonality is not answerable in those terms at all, because it reveals that the claim and the allocation were never operating on the same axis. Contracts followed industrial capacity, transport, and existing agglomeration; installations followed terrain, climate, and airspace; deaths followed the draft and the fortunes of particular divisions. That these three geographies failed to coincide is not a scandal of administration. It is what one should expect when three allocation processes optimize over criteria that have nothing to do with one another. The language of shared sacrifice did not conceal a policy of unequal reward; it supplied a moral vocabulary for a distribution that no one was choosing along that dimension.

What the language did conceal is the *variance*. Nationally, 42.9 Army and Army Air Forces enlistees died per thousand who served. Across counties the ninetieth percentile of that rate was 3.58 times the tenth. Some 1,553 counties—more than half of those that lost men—received no major war contract of any kind. A national idiom of common contribution had to survive contact with a distribution in which a county's experience of the war was substantially a matter of which of four unrelated allocations happened to reach it. The rhetoric was not false about the fact of sacrifice. It was silent about the geography of it, and that silence was doing work: it converted a spatially arbitrary distribution into a nationally shared condition, which is precisely the conversion required to sustain consent.

There is a further asymmetry the postwar record makes visible and the wartime language could not. Federal money that arrived as industrial plant and federal money that arrived as a military installation were, in 1943, equally legible as the government's presence, equally describable as the community's contribution to the war. By 1970 they had left different structures behind—the one having added manufacturing employment and left tenure where it found it, the other having added employment of every other kind and a measured home-ownership rate lower by seven and a half points, some part of which is the tenure a standing garrison imposes. The wartime vocabulary had no way to mark that difference, because the difference had not happened yet and because the vocabulary was designed to make forms of contribution equivalent rather than to distinguish among them. The compression of geography into a national idiom was thus not only a compression of who sacrificed. It was a compression of what kinds of federal presence a community was receiving, and therefore of what it would still have when the war ended.

This is why the paper measures correspondence rather than compensation. Asking whether investment repaid sacrifice would require a rate at which capital substitutes for the dead, and no such rate exists. Asking whether the two distributions occupied the same space is

answerable, and the answer is what gives the rhetorical claim its historical interest: per-capita rank correlations of -0.026 , $+0.013$ and -0.054 against prewar loadings of 0.23 to 0.69, and a coverage asymmetry in which 98.2 percent of counties recorded a death and 48.5 percent received a major contract.²² Shared sacrifice need not be dismissed as propaganda because sacrifice was unequally distributed. Its political function may have depended on exactly the suppression of geographic detail that the county data restores.

7 Conclusion

World War II redistributed both people and capital across the United States, and scholarship has studied those redistributions separately. Placing them in one county panel changes what can be said about either.

Three findings follow. First, the wartime state made not one allocation but three, and they do not coincide. Procurement contracts, publicly financed industrial plant, and military installations occupy geographies whose pairwise overlap runs from 0.22 to 0.59. The two capital programs are the weakly related pair, correlating at $\rho = 0.31$; contracts and plant, both of which followed existing industry, are the exception at $\rho = 0.62$. Of 2,973 counties, 118 received a large public plant and 56 a top-decile installation, and only two received both — against 2.2 expected if the two programs had been allocated independently of one another.

Second, none of the three tracked sacrifice. Measured per resident on the same 3,073 counties, contract dollars correlate with the 1940 manufacturing share at $+0.693$ and with fatal burden at -0.026 ; the other two allocations load on prewar characteristics at $+0.23$ to $+0.51$ and on burden at $+0.013$ and -0.054 . The wartime allocation function had a large weight on prewar industrial capacity and no meaningful weight on human loss, and at this sample size these exclude any association above 0.09 in absolute value. Coverage says the same: 98.2 percent of counties recorded a death while 48.5 percent received any major contract, and 1,553 counties lost men and received no major war contract at all. The rank correlations turn negative when deaths are conditioned on enlistment. Adjusting for the composition of the men a county sent — race, education, birth year and volunteer status from 7.2 million individual enlistment records, and test score from the 5 percent of them that carry one — leaves all three negative, and makes two of the three more so. The war state was not hostile to the places that bore the most sacrifice. With respect to sacrifice it was indifferent.

²²Not the overlap coefficient, which Section 5 shows is dominated by the concentration of contracts relative to population — 85 percent of the shortfall — and is therefore largely a restatement of a fact established before this paper.

Third, the investment that did arrive left durable and divergent structures. A large new publicly financed plant raised county population, manufacturing employment and family income through 1970, and raised manufacturing employment faster than employment as a whole. A military installation raised population and total employment by more, did not measurably raise manufacturing employment, and lowered measured homeownership by seven and a half points. The two allocations did not have opposite effects on manufacturing; one of them acted on manufacturing and the other acted on everything around it. Both designs pass placebo tests on pre-treatment outcomes, and balance on all but a few prewar covariates. The plant design leaves two of nine standardized differences just past the conventional 0.10 threshold, latitude and the 1940 Black population share; the military design leaves one, 1930–40 population growth at 0.106. All three enter the outcome regressions as controls.

What is and is not claimed

The causal claims are narrower than the estimates alone would license, and the boundary should be stated plainly.

For the plant design, the population, manufacturing and family-income effects survive the most demanding available comparison — treated counties against their own untreated neighbors, sharing a labor market, a climate, and a state. **There is no homeownership effect to claim.** The point estimate is +0.94 points against matched controls ($p = 0.12$) and +0.40 against adjacent counties ($p = 0.48$), and the continuous specification of Table 7 puts it at zero. A war plant changed how many people lived in a county and what they did for a living; it did not change how many of them owned their homes a generation later. The surviving income effect is also smaller under the neighbor comparison — 4.6 percent against 7.3 percent — so roughly a third of the national-pool estimate is regional rather than local.

For the military design the estimates are stable across five treatment definitions with a monotone dose-response in two of the three outcomes, and randomization inference on 500 permutations returns $p \leq 0.004$. But the treated group is 52 counties, and no neighbor comparison is available at that size.

Nor does this paper claim to have identified wartime investment causally in general. The natural instrument — the 1938 Industrial Mobilization Plan — passes the first-stage F test, is weak on partial R^2 , and fails its exclusion restriction outright, because the Munitions Board allocated plants that were already industrially capable in places that were already urban. That failure is a property of the setting and not of one construction. Of nine ways of building the instrument from that source, from every allocated establishment to the aeronautical branch alone, none clears relevance and exclusion together. Nor does a shift-share assembled

from a different archive: it is the stronger instrument, at $F = 45.1$ against 19.7, and fails the same three placebos at $p < 0.001$. The reason is structural. The industries that dominated war spending were built where they had not existed — 27.9 percent of aircraft dollars and 77.2 percent of ammunition dollars went to counties with no prewar plant of that kind — so any instrument built from prewar capacity can identify only the procurement that ran through existing plants, which is the procurement prewar industrial structure explains. Section 5.9 reports that result rather than the estimates the instrument produces. What remains is a matched comparison for two specific treatments, and a descriptive result about alignment that requires no identification at all.

Implication

The rhetorical claim examined in Section 6 was a claim about distribution, made while the distribution was being decided. The measured geography does not show that claim to have been inverted — investment did not flow away from the places that lost most. It shows something less dramatic and harder to answer: the two were decided on unrelated criteria. Contracts followed industrial capacity and transport; installations followed terrain, climate, and airspace; deaths followed the draft and the fortunes of particular divisions. A language of common contribution had to carry a distribution that no one was choosing along that dimension, and by 1970 the counties that had received one kind of federal capital and the counties that had received another were different kinds of place — while the counties that had furnished the dead were not, on that account, any kind of place at all.

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A Construction, Coverage, and Robustness

This appendix reports construction detail, coverage diagnostics, and the robustness and falsification tests referenced in the text.

Table A1: Place-to-county assignment by match stage

Match stage	Records	% records	% dollars
<i>War supply contracts</i> ($N = 190,693$)			
Exact	159,129	83.4	83.1
Alias	23,911	12.5	15.2
Fuzzy	439	0.2	0.1
Unmatched	7,214	3.8	1.5
<i>War facilities (assignable)</i> ($N = 14,501$)			
Exact	12,173	83.9	70.8
Alias	1,091	7.5	26.3
Fuzzy	216	1.5	0.6
Unmatched	1,021	7.0	2.2
Source records no place	180	—	3.7

Notes: Stages are applied in order and are mutually exclusive. Aliases are curated historical identifications of boroughs, company towns, and unincorporated plant sites; fuzzy matching is within-state at a 0.88 similarity threshold with an explicit blocklist for near-misses that cross county lines. Facilities percentages are of publicly financed dollars.

Table A2: County characteristics by type of war facility received

County type	N	Burden	Pop. 1940	Urban	Mfg. 1940	Mfg. 1970	Income 1970
Base only	349.0	2.19	23,858	0.340	0.016	14.4	7,602
Both	275.0	2.27	88,415	0.616	0.051	23.3	9,499
Neither	2,063.0	2.31	14,227	0.000	0.007	19.7	6,820
Plant only	386.0	2.36	35,558	0.397	0.052	33.0	8,769

Notes: Medians. Burden is Army/AAF deaths per 1,000 1940 residents.

Table A3: The twenty largest contract-receiving counties

County	Contracts (\$m)	Plant (\$m)	Military (\$m)	Deaths	B_i
WAYNE	11,218	615	20	1,598	0.79
LOS ANGELES	9,716	415	108	6,385	2.29
COOK	8,421	727	27	8,720	2.15
CUYAHOGA	4,476	214	6	2,712	2.23
ERIE	4,217	233	1	2,059	2.58
BALTIMORE	4,128	141	11	365	0.36
NEW YORK	3,639	24	47	4,654	2.46
HUDSON	3,519	110	46	1,581	2.42
HARTFORD	3,468	119	8	1,020	2.27
HAMILTON	2,704	177	5	1,327	2.13
MARION	2,567	170	7	1,045	2.27
PHILADELPHIA	2,557	221	92	4,308	2.23
PASSAIC	2,523	44	0	744	2.41
NASSAU	2,522	101	10	772	1.90
KING	2,235	55	77	1,090	2.16
KINGS	2,162	223	35	5,709	2.12
SAN DIEGO	2,036	44	215	763	2.64
ST LOUIS	1,978	222	17	135	0.12
SCHENECTADY	1,964	20	21	278	2.27
ST JOSEPH	1,907	67	0	469	2.90

Notes: Dollar figures in millions.

Table A4: Pre-trend falsification: wartime exposure against pre-1940 outcomes

Placebo outcome	W^C	W^F	W^M	B	N
log pop growth 1920->1930	0.0004	0.0044	0.0008	-0.0004	3,041
log pop growth 1930->1940	0.0033***	-0.0002	0.0047***	-0.0012***	3,073
mfg share 1940 (pre-war level)	0.0046***	0.0010*	-0.0014***	0.0000	3,073

Notes: Wartime exposure was assigned 1940–45 and should not predict earlier outcomes. Controls and fixed effects as in Table 7, excluding the dependent variable from the control set.

Table A5: The interactions of equation (10)

Outcome, 1970	θ_C	(SE)	θ_F	(SE)	N
log population 1970	0.00008	(0.00012)	0.00017	(0.00012)	2,971
manufacturing share 1970	0.01154***	(0.00270)	0.00355*	(0.00214)	2,973
log family income 1970	-0.00002	(0.00008)	0.00003	(0.00008)	2,973
homeownership rate 1970	0.00227	(0.00140)	0.00219	(0.00143)	2,973

Notes: θ_C and θ_F are the coefficients on $W_i^C \times B_i^A$ and $W_i^F \times B_i^A$ in equation (10), estimated with the full prewar control vector and state fixed effects on the counties passing the validity screen for B_i^A . Standard errors clustered on state. *, ** and *** denote significance at the 10, 5 and 1 percent levels.

Table A6: Alternative denominators for fatal burden

Denominator for B_i	Coefficient	p	N
per 1,000 residents	-0.00226	0.382	3,065
per 1,000 males	-0.00117	0.368	3,065
per 1,000 men in the labor force	-0.00077	0.294	3,065
battle deaths only	-0.00502	0.147	3,065

Notes: Each row re-estimates the 1970 log-population specification of Table 7 with a different construction of B_i .

Table A7: Conley spatial standard errors against state clustering

Variable	Coefficient	Clustered SE	Conley SE	t (Conley)
<i>log population 1970</i>				
W^C	0.0346	0.0034	0.0042	8.25
W^F	0.0197	0.0046	0.0039	5.00
W^M	0.0493	0.0047	0.0047	10.56
B	-0.0023	0.0026	0.0025	-0.90
<i>mfg share 1970</i>				
W^C	0.3325	0.1016	0.0977	3.40
W^F	0.2851	0.0829	0.0873	3.26
W^M	-0.5945	0.1108	0.0956	-6.22
B	-0.0798	0.0961	0.0669	-1.19

Notes: Spatial HAC standard errors following Conley (1999), with a uniform kernel at 200 km computed from county centroids.

Table A8: Excluding the twenty largest contract recipients

Variable	Full sample	Excl. top 20
<i>log pop 1970</i>		
W^C	0.0346	0.0336
W^F	0.0197	0.0200
W^M	0.0493	0.0491
B	-0.0023	-0.0024
<i>mfg share 1970</i>		
W^C	0.3325	0.3180
W^F	0.2851	0.2921
W^M	-0.5945	-0.5987
B	-0.0798	-0.0786

Table A9: Covariate balance before and after matching

Covariate	Raw SMD	Matched SMD
Log population 1940	1.450	-0.028
Urban share 1940	1.262	0.027
Black population share 1940	0.093	0.107
Manufacturing share 1940	0.627	-0.097
Log population growth 1930–40	0.450	0.065
Illiteracy rate 1930	-0.168	0.046
Farm share 1930	-0.165	-0.048
Latitude	-0.147	-0.124
Longitude	-0.047	0.077
Imbalanced ($ \text{SMD} > 0.10$)	7 of 9	2 of 9

Notes: Standardized mean difference between treated and control counties. Values above 0.10 in absolute value are conventionally treated as imbalance. Matched differences are weighted by the matching weights.

Table A10: Placebo: matched design applied to pre-treatment outcomes

Pre-treatment outcome	Coefficient	p	N
log pop growth 1930–40	0.0354	0.664	364
mfg share 1940	-0.0211	0.805	364
homeownership 1940	1.2687	0.107	364

Notes: A credible design should show no treatment effect on outcomes determined before the plants were sited.

Table A11: Robustness of the matched design

Specification	Log pop. 1970	Mfg. share 1970	Log income 1970	Own rate 1970
<i>Matching parameters</i>				
1:1, caliper 0.2sd	0.211	4.047	0.079	0.791
3:1, caliper 0.2sd (baseline)	0.209	3.338	0.070	0.942
5:1, caliper 0.2sd	0.193	3.419	0.070	0.766
3:1, caliper 0.1sd	0.209	3.623	0.071	0.916
3:1, caliper 0.5sd	0.201	3.447	0.073	0.993
<i>Sample restrictions</i>				
pop $\geq 10,000$	0.197	3.634	0.081	1.184
any 1940 manufacturing	0.217	3.156	0.085	0.981
<i>Treated against their own untreated neighbors</i>				
Coefficient	0.180***	1.628**	0.045**	0.397
p-value	0.002	0.022	0.012	0.477

Notes: Each row re-estimates the four 1970 outcomes of Table 8 under a different matching rule or sample restriction. The final panel restricts the control pool to the 475 untreated counties that directly border a treated county, then applies the same propensity match within that pool. Metropolitan status is not varied: the estimation sample already excludes the hundred largest manufacturing counties of 1940. Stars on the final panel denote significance at the 10, 5 and 1 percent levels.

Table A12: Selection on unobservables required to overturn each estimate

Outcome	β , short	β , long	β , matched	R^2	δ at $1.3R^2$	δ at $R^2 = 1$
<i>Large public plant</i>						
log population 1970	1.7783	0.2289	0.2092	0.881	0.98	0.98
manufacturing share 1970	6.2108	3.2037	3.3378	0.540	3.50	1.23
log family income 1970	0.2407	0.0789	0.0704	0.509	1.51	0.47
homeownership 1970	-2.0764	1.8373	0.9416	0.366	-1.55	-0.27
log manufacturing employment 1970	2.2763	0.4152	0.3878	0.807	0.86	0.86
<i>Top-decile military installation</i>						
log population 1970	0.3668	0.5937	0.5509	0.884	-19.92	-19.92
manufacturing share 1970	-7.5620	-4.4418	-4.7201	0.540	4.70	1.66
log family income 1970	0.1043	0.1192	0.1051	0.509	-26.43	-8.22
homeownership 1970	-8.7683	-7.7274	-7.4890	0.383	23.25	4.32

Notes: δ following Oster (2019) is the ratio of selection on unobservables to selection on observables that would move the treatment coefficient to zero, comparing a short regression with no controls to a long one with the full prewar vector. Both are estimated on the full estimation frame without matching or weights, since δ measures how far the coefficient travels when the observables are added and matching removes that movement by construction; the matched estimate is printed beside the long coefficient for comparison. $|\delta| > 1$ is the conventional robustness threshold. A negative δ means the prewar vector moves the coefficient away from zero, so observed selection runs against the estimated effect. $R^2_{\max}$ is set at 1.3 times the controlled R^2 , following Oster, and again at one.

Table A13: Placebo tests, military-installation design

Pre-treatment outcome	Coefficient	p
log pop growth 1930-40	0.0862	0.661
mfg share 1940	-0.0689	0.625
homeownership 1940	0.5290	0.697

Table A14: Military-installation design: threshold sensitivity and randomization inference

Treatment definition	Treated	Log pop. 1970	Mfg. share 1970	Own rate 1970
top 5% of per-capita military \$	27	0.867***	-9.092***	-9.252***
top 10% (baseline)	52	0.551***	-4.720***	-7.489***
top 20%	99	0.535***	-4.306***	-5.610***
top 30%	153	0.385***	-4.611***	-4.690***
Military spending $\geq$ \$10M (absolute)	183	0.375***	-4.178***	-4.290***

Randomization inference, 500 permutations of treatment

log pop: observed 0.551	permutation $p = 0.000$, null s.d. 0.063
mfg share: observed -4.720	permutation $p = 0.004$, null s.d. 1.422
own rate: observed -7.489	permutation $p = 0.000$, null s.d. 0.976

Notes: Each row is a separate matched design on its own propensity score. Permutation p -values are the share of 500 random reassignments of treatment producing an absolute effect at least as large as the observed one.

Table A15: First-stage strength of the 1938 Industrial Mobilization Plan instruments

Specification		F	N	Verdict
contracts	IMP facilities	23.5	3,073	strong
contracts	IMP by branch (4)	8.8	3,073	weak
war plant	IMP industrial	63.6	3,073	strong
war plant	IMP by branch (4)	13.6	3,073	strong
military	prewar bases/estabs	18.9	3,073	strong

Notes: Each row instruments the named endogenous regressor with the named instrument set, conditional on the full prewar control vector and state fixed effects. F is the joint test on the excluded instruments, with standard errors clustered on state. Conventional practice treats $F < 10$ as weak.

Table A16: Single-endogenous IV estimates against OLS

Outcome	OLS	IV	(SE)	Partial R^2	F
<i>war plant, instrumented by IMP industrial</i>					
log population 1970	0.0185	0.0055	(0.0261)	0.0398	70.3
manufacturing share 1970	0.1753	-1.7365**	(0.8337)	0.0397	69.8
log family income 1970	-0.0324	-0.4301***	(0.0791)	0.0397	69.8
homeownership 1970	-0.3269	-5.2921***	(1.2178)	0.0397	69.8
<i>contracts, instrumented by IMP facilities</i>					
log population 1970	0.0339	0.0116	(0.0447)	0.0101	15.3
manufacturing share 1970	0.2872	-3.0800	(2.0066)	0.0100	15.1
log family income 1970	-0.0095	-0.7036***	(0.2393)	0.0100	15.1
homeownership 1970	-0.2207	-8.9496**	(3.6313)	0.0100	15.1
<i>military, instrumented by prewar bases</i>					
log population 1970	0.0491	-0.0252	(0.0438)	0.0103	18.9
manufacturing share 1970	-0.6085	-3.8888***	(1.2019)	0.0103	19.0
log family income 1970	0.0029	-0.4233***	(0.1286)	0.0103	19.0
homeownership 1970	-0.3852	-7.4356***	(2.2856)	0.0103	19.0

Notes: One treatment is instrumented at a time; the other two enter as exogenous covariates. Partial R^2 is that of the excluded instrument after controls and state effects. The near-identical large negative coefficients across three different instruments indicate a common confounder rather than three causal effects.

Table A17: Exclusion-restriction placebo: instruments against pre-war outcomes

Pre-war outcome	IMP industrial	p	Prewar bases	p
log population growth 1930-40	0.0000	0.719	0.0000	0.889
manufacturing share 1940	0.0135***	0.000	-0.0074***	0.000
log manufacturing payroll 1940	-0.0787**	0.012	-0.0953**	0.024
urban share 1940	0.0896***	0.000	0.1340***	0.000

Notes: A valid instrument should not predict outcomes determined before the treatment. Both instruments predict the 1940 manufacturing and urban shares, though neither predicts 1930–40 population growth: they mark prewar industrial *level*, not prewar *trend*.

Table A18: Every construction of the 1938 instrument, relevance against exclusion

Construction of the instrument	Endog.	Relevance		Placebo p : instrument against a pre-war outcome			
		F	Partial R^2	Mfg. share	Urban share	Log pop.	Pop. growth
All allocated establishments, every branch	W^C	37.3	0.027	0.000***	0.000***	0.000***	0.002***
Allocated facilities (baseline)	W^C	23.5	0.018	0.000***	0.000***	0.000***	0.000***
Facilities, machine tool, aero, optics, steel	W^C	23.6	0.018	0.000***	0.000***	0.000***	0.000***
Any allocated facility (extensive margin)	W^C	59.1	0.055	0.000***	0.000***	0.000***	0.885
Machine-tool branch only	W^C	3.9	0.001	0.000***	0.785	0.000***	0.000***
Aeronautical branch only	W^F	10.7	0.003	0.162	0.212	0.000***	0.000***
Aeronautical and optical branches only	W^F	13.3	0.003	0.197	0.263	0.000***	0.000***
Prewar Army and Navy sites	W^M	18.9	0.011	0.000***	0.002***	0.000***	0.001***
Any prewar Army or Navy site	W^M	24.9	0.014	0.008***	0.000***	0.000***	0.038**

Notes: Each row is a separate construction of the instrument, entering in inverse hyperbolic sine. F is the first-stage test on the excluded instrument and the partial R^2 is that of the instrument after the full prewar control vector and state fixed effects. The four right-hand columns regress a pre-war outcome on the instrument with the same controls, excluding that outcome from the control set; a valid instrument should not reject. A construction is usable only if $F \geq 10$, partial $R^2 \geq 0.05$, and no placebo rejects at five percent. None of the nine satisfies all three. Standard errors clustered on state throughout; *, **, *** denote 10, 5 and 1 percent.

Table A19: A shift-share instrument against the 1938 plan

Instrument	F	Partial R^2	Mfg. share	Urban share	Log pop.	Pop. growth
shift-share, all counties	45.1	0.0385	0.000***	0.000***	0.000***	0.034**
shift-share, county left out	45.0	0.0385	0.000***	0.000***	0.000***	0.034**
shift-share, state left out	44.7	0.0383	0.000***	0.000***	0.000***	0.032**
IMP allocated facilities	19.7	0.0141	0.000***	0.000***	0.000***	0.001***

Notes: The shift-share interacts each county's share of the national 1940 stock of an industry with national contract dollars placed in the matching product class, and enters in inverse hyperbolic sine per 1940 resident. The leave-out rows remove the county's own contracts, and its whole state's, from the national total. F and the partial R^2 are for the excluded instrument in the equation for W^C after the full prewar control vector and state fixed effects; the four right-hand columns report p values from regressing a pre-war outcome on the instrument with the same controls, excluding that outcome from the control set. The Munitions Board row is the baseline of Table A15, re-estimated on the identical sample. Standard errors clustered on state.

Table A20: Where the shift-share gets its identification, and where the war spent its money

Product class	\$bn	Counties with prewar capacity	Counties receiving	% of \$ to a county with none	Partial R^2 of this class alone
Aircraft	51.6	43	317	27.9	0.0001
Ships	25.6	167	383	19.9	0.0049
Ammunition	14.1	47	505	77.2	0.0000
Electrical and radio [†]	13.1	703	376	20.5	0.0034
Small arms and ordnance	12.1	15	331	78.7	0.0002
Motor vehicles	10.5	423	375	6.3	0.0167
Textiles and clothing	8.3	926	831	14.2	0.0366
Machine tools and machinery [†]	7.9	2,998	478	5.3	0.0316
Petroleum	5.4	647	266	14.1	0.0100
Tanks and combat vehicles	5.0	100	191	17.0	0.0016
Iron and steel	3.3	560	361	5.9	0.0146
Instruments and optics	2.4	67	157	43.2	0.0007
Explosives	1.6	60	143	89.0	0.0002
Chemicals	1.3	1,031	306	24.1	0.0096
Leather and footwear	1.0	427	214	10.6	0.0028
Rubber	1.0	143	194	23.0	0.0004
Lumber and paper	0.9	1,907	569	10.6	0.0013
Railroad equipment	0.8	377	103	6.1	0.0015
Nonferrous metals	0.6	459	162	31.9	0.0041
Stone, clay and glass	0.2	1,109	188	6.8	0.0055
Food [†]	0.0	2,936	63	13.1	0.0015

Notes: Product classes are assigned from the contract file’s free-text product description by the keyword classifier described in Section 5.9; 95.1 percent of contract dollars are classified. The prewar column counts 1940 county manufacturing establishments in the matching industry, from the Jaworski (2017) archive, except for the three rows marked [†], where the archive carries no 1940 establishment count and the share is built from 1930 county employment instead. Iron and steel therefore enters twice, under both definitions, so those two rows are not independent. Dollars are the geocoded base, which is the base the greenfield share is computed on; the classified national total is \$169.1 billion. The final column is the partial R^2 of that class’s share instrument entered alone, after the full prewar control vector and state fixed effects. Classes are ordered by wartime dollars. The four largest — aircraft, ships, ammunition and electrical equipment — carry 62.2 percent of classified spending and a combined partial R^2 of 0.008, while the four that carry 68 percent of the summed identification — textiles, machine tools, motor vehicles and iron and steel — carry 18.3 percent of the dollars.

Table A21: The allocation function, re-estimated for each year of the war

Award year	1940	1941	1942	1943	1944	1945
Dollars placed (\$bn)	10.5	17.7	58.5	45.5	33.2	12.7
Counties receiving (%)	15.0	23.4	36.6	40.1	37.6	35.0
ρ (1940 mfg. share)	0.497	0.514	0.584	0.483	0.579	0.588
B_i per resident	0.00400***	0.00550***	0.00761***	0.00806***	0.00787***	0.00736***
p	0.000	0.000	0.000	0.000	0.000	0.000
B_i^A per enlistee	0.00031	0.00110	-0.00120	-0.00005	-0.00106	-0.00027
p	0.691	0.275	0.381	0.969	0.443	0.832

Notes: Contracts are assigned to a year by their award month, which every record carries. This exhibit uses only that month and the county assignment of Section 5; it does not use the product classifier of Section 5.9. The rank correlation is computed on the fixed set of 1,490 counties receiving a contract at some point during the war, so that the rising share of counties with any contract does not drive it. The two burden rows are coefficients on fatal burden in a regression of the inverse hyperbolic sine of that year's contract dollars per resident on the full prewar control vector and state fixed effects; B_i^A is defined on the 2,976 counties passing the validity screen. Standard errors clustered on state.